



Full length articles

The distributional effects of carbon pricing across countries^{☆,☆☆}Mathilde Le Moigne^a, Simon Lepot^a, Marcos Ritel^{b,*}, Dora Simon^c^a University of Zurich, Switzerland^b Kuehne Logistics University, Germany^c University of Stavanger, Norway

ARTICLE INFO

Dataset link: [The Distributional Effects of Carbon Pricing - Replication Package \(Original data\)](#)

Keywords:

International trade

Climate policy

Climate change

ABSTRACT

We use a quantitative international trade model with climate policies to explore the distributional effects of carbon pricing across countries. Our analysis addresses two key questions facing global climate action: which countries bear the greatest burden of climate policies, and how these policies can be designed to ensure fairness. We present three main findings. First, efficient climate policies that disregard distributional concerns significantly exacerbate between-country inequality. Second, equity can be achieved alongside efficiency when climate policies are complemented by economically feasible international transfers, either equalizing carbon tax costs or accounting for historical emissions, with minimal economic impact on high-income countries. Third, carbon tax schemes with heterogeneous pricing – featuring lower rates for low- and middle-income countries – do not necessarily result in fairer outcomes.

1. Introduction

Climate change is one of the defining global challenges of our time. It is driven by greenhouse gas (GHG) emissions, an externality whose damages are distributed unevenly across countries. Low- and middle-income economies are substantially more vulnerable to climate hazards,¹ despite having contributed relatively little to the historical accumulation of atmospheric GHGs. This asymmetry has long motivated the Global South to call for differentiated responsibilities in international climate negotiations, most notably articulated in the principle of Common but Differentiated Responsibilities (CBDR) codified in the 1992 UNFCCC.² Yet, despite its centrality to climate diplomacy, CBDR has rarely been translated into concrete global policy, and disagreements around differentiation remain a key obstacle to progress in climate governance.

In this paper, we provide the first comprehensive quantitative assessment of the distributional consequences of global carbon policies across countries. Using a state-of-the-art general-equilibrium trade model that embeds climate externalities, we quantify the extent to which carbon pricing creates or reduces inequalities between the Global North and South. We view this exercise as an

[☆] This article is part of a Special issue entitled: 'Trade and Uneven Development' published in Journal of International Economics.

^{☆☆} This work is supported by the Kuehne Center for Sustainable Trade and Logistics at the University of Zurich. We thank Luca Poll for outstanding research assistance. We also thank Ralph Ossa, Ahmad Lashkaripour and two anonymous referees for invaluable comments and suggestions. This paper was presented at several venues as part of the project Sustainable Globalization and is the scientific basis for the Kuehne Impact Series 02-23 "The Distributional Effects of Carbon Pricing: A Global View of Common but Differentiated Responsibilities". We thank participants at various seminars and conferences, especially the World Bank Trade and Uneven Development conference. All errors remain our own.

* Corresponding author.

E-mail address: marcos.ritel@klu.org (M. Ritel).

¹ See IPCC Working Group II (Adler et al., 2022).

² The United Nations Framework Convention on Climate Change (UNFCCC) is the foundational international treaty adopted at the 1992 Rio Earth Summit, establishing the institutional framework and guiding principles for global climate cooperation.

important benchmark: even if a globally coordinated carbon tax is politically difficult to implement, understanding its distributional implications clarifies what fairness would require and provides a reference point against which more realistic policy proposals can be evaluated.

Our analysis proceeds in three steps. We begin by establishing a benchmark that asks what the consequences of climate policies that disregard distributional concerns are. This involves the adoption of a single Pigouvian-style global price for CO₂ equivalents (CO₂e), a popular proposal among policymakers and experts.^{3,4} We quantify its global economic and environmental effects and document how they differ across income groups.

We then ask whether a global carbon tax can be compatible with principles of equity. A key insight from the public-finance literature is that efficiency and equity can be reconciled in the presence of international lump-sum transfers (Sandmo, 2006). Motivated by this result, we compute transfers that either equalize the contemporary economic costs of carbon taxation or align those costs with countries' historical contributions to cumulative global emissions. These counterfactuals allow us to quantify the scale and distribution of transfers required to make a uniform global carbon tax consistent with different notions of fairness, and to compare these magnitudes with actual climate-finance flows.

Finally, we evaluate a prominent alternative to uniform pricing: heterogeneous carbon prices indexed to countries' income levels. Such schemes are often viewed as more politically feasible and potentially fairer for low- and middle-income economies. We focus on the International Carbon Price Floor (ICPF) proposed by the International Monetary Fund (IMF) (Parry et al., 2021), and examine several variants ranging from the IMF's "restricted club" of major emitters to a version in which all countries participate according to their income classification.

An important aspect of our approach is the choice of equity criterion. Although one could design transfers that equalize total welfare changes, including the long-run benefits of avoided climate damages, we deliberately focus on equalizing contemporaneous economic costs. Transfers are meant to offset the immediate and politically salient burden of climate action, whereas the welfare gains from mitigation materialize decades in the future and rest on highly uncertain estimates of the social cost of carbon. In our view, equalizing present-day losses therefore offers a more transparent and policy-relevant benchmark for evaluating distributional fairness. Still, we note that equalizing total welfare changes would generally reduce the scale of transfers relative to our baseline, especially for low- and middle-income countries that benefit the most from avoided climate damages, as we document in this paper.

Our investigation builds on a general-equilibrium international trade model with multiple countries and sectors, incorporating a rich description of the world economy that includes trade in final and intermediate goods, sectoral heterogeneity, and input–output linkages (Armington, 1969; Caliendo and Parro, 2015). It features extraction of three fossil fuels – oil, coal and gas – which, together with agricultural production, generates a climate externality.⁵

The model covers 64 countries and regions and 44 tradable and non-tradable sectors. We leverage rich data on CO₂, CH₄, and N₂O emissions, which together account for nearly all GHG emissions. Governments in the model can implement a carbon tax, with the tax revenues rebated either to the country of production or consumption of final and intermediate goods. Our benchmark policy is a 100 USD/tCO₂e carbon tax, which is close to existing applied carbon taxes. Nonetheless, all conclusions of our analysis are robust to setting this policy at different levels.⁶

Our analysis delivers three main insights. First, we show that a uniform carbon tax strongly exacerbates between-country inequality. While it is widely acknowledged that low- and middle-income countries are most vulnerable to climate change (Adler et al., 2022), our results demonstrate that they also shoulder the largest present-day costs of climate action. Although a 100 USD/tCO₂e carbon tax is effective on average, reducing global emissions by 52% at a modest real-income loss of 0.5%, this average conceals substantial heterogeneity: low- and middle-income economies face real-income declines of up to 8%, whereas many high-income countries experience gains. Incorporating avoided climate damages makes mitigation more attractive overall, particularly for countries facing severe future climate losses, but it does not overturn the basic pattern for all low-income economies, as many still experience net welfare losses even after accounting for avoided damages.

These results stem from a heavier carbon tax burden in the Global South, driven by two main factors. For starters, the Global South tends to specialize in more polluting industries. More specifically, low and middle-income countries are often more upstream in international value chains, providing raw materials that are notably more carbon-intensive than other sectors. This specialization also results in a more carbon-intensive consumption basket, as domestic production constitutes a large portion of domestic expenditure. Moreover, the Global South tends to produce using more polluting technologies, thus emitting more in any given sector per unit

³ For instance, Nobel laureate Jean Tirole recently expressed this view arguing that: "since the emission of a ton of greenhouse gases causes the same environmental damage, wherever, whenever and however it is emitted, a single global price for CO₂ should guide public and private agents in their investment, production and consumption decisions" (Tirole, 2016).

⁴ A uniform carbon tax is the optimal policy under global cooperation in a perfectly competitive environment and in the absence of distributional considerations. See Farrokhi and Lashkaripour (2025) for a characterization of optimal climate policies and their consequences in a general equilibrium environment.

⁵ Following recent work, we model fossil-fuel extraction as dependent on a fixed factor, which gives rise to upward-sloping supply curves (Farrokhi and Lashkaripour, 2025; Garcia-Lembergman et al., 2025). We allow fossil-fuel prices to vary across countries, consistent with observed price dispersion in oil, coal, and natural gas.

⁶ Note that the magnitude of the optimal global carbon tax is defined by the social cost of carbon. As noted by Blanchard et al. (2023) "there exists no consensus among economists about how large this Pigouvian carbon price should be". Our calibration is also close to Farrokhi and Lashkaripour (2025), who set the social cost of carbon in 2014 at 156 USD/tCO₂ based on estimates provided by the United States Environmental Protection Agency.

produced. Consequently, as carbon taxes cause the environmental cost of carbon to be reflected in relative prices, economic activity is reallocated towards greener sectors predominantly found in high-income markets.

Given that some countries experience real-income gains while others face losses, there is a natural role for international transfers to equalize the economic burden of climate action. Our second main result shows that feasible transfers from the Global North to the Global South can fully offset these disparities while preserving the emissions reductions generated by a uniform global carbon tax. Depending on the tax regime and the allocation principle, North–South transfers of USD 107–169 billion are sufficient to equalize the economic costs of climate action. On a per-capita basis, this corresponds to average transfers of roughly USD 89–200 from high-income to low- and middle-income countries—amounts that would eliminate the unequal burden of climate action while maintaining the same level of global emissions reduction.

We derive this result by examining two transfer schemes, both of which generate flows from high- to low- and middle-income countries. The first equalizes real-income losses across countries. Under the consumer-tax regime, European economies and the United States are the main contributors, while large low- and middle-income economies – such as Kazakhstan, Russia, South Africa, and several Southeast Asian countries – receive transfers. Under the producer-tax regime, high-income fossil-fuel producers become the principal contributors, as origin-based taxation allows them to retain significant carbon-tax revenue. Required transfers range from USD 231 bn (consumer tax) to USD 391 bn (producer tax), roughly tripling current climate-finance commitments but implying manageable per-capita contributions for most high-income countries.

The second scheme assigns each country a burden proportional to its historical contribution to cumulative emissions. The total transfers required – USD 237 bn under consumer taxation and USD 201 bn under producer taxation – are similar to or smaller than those under the equal-cost scheme, with high-income economies again financing transfers to most low- and middle-income countries. Because burdens are tied to historical responsibility, North–South transfers rise modestly relative to the equal-cost baseline, and the distinction between consumer- and producer-tax regimes becomes largely irrelevant, yielding almost identical global transfer patterns.

Our third result shows that although heterogeneous carbon pricing may be more politically feasible and reduce free-riding incentives, it does not yield fairer outcomes relative to a global carbon tax with redistribution. We evaluate the IMF's International Carbon Price Floor (ICPF) across several configurations—from the Fund's proposed “restricted club” to a version in which all countries participate with income-differentiated price floors.⁷ Across all variants, the ICPF fails to achieve both lower transfer needs and higher efficiency at the same time when compared with a uniform global carbon tax accompanied by transfers. Even when all countries participate, the modest reduction in transfers comes with substantially larger global real-income losses. More restricted versions perform worse on fairness, requiring higher North–South transfers than the uniform-tax benchmark. Overall, heterogeneous carbon pricing, as reflected in current proposals, does not inherently improve equity and can exacerbate cross-country inequality.

This paper contributes to several strands of the literature. Our study adds to the extensive literature on international trade, trade policy and greenhouse gas emissions. Earlier studies focused on small-scale models, often analyzing the effects of unilateral climate policy on trade, notably on leakage (e.g. Markusen (1975), Hoel (1996), Copeland (1996)).⁸

More recently, several studies have begun to employ quantitative trade models to analyze environmental and trade policies. Closer to our analysis of global carbon policies is Farrokhi and Lashkaripour (2025), who examine the effectiveness of optimal environmental and trade instruments – specifically carbon border taxes and climate clubs with trade penalties – as mechanisms to sustain global climate cooperation and reduce emissions. Garcia-Lembergman et al. (2025) advance this agenda by using a quantitative model to estimate the supply elasticity of fossil fuels and quantify the global effects of trade shocks and carbon taxes. Caliendo et al. (2024) and Farrokhi et al. (2025) further investigate the interplay between trade integration and environmental externalities. We add to this literature by shifting the focus to the distributional and socioeconomic impacts of global climate policies on low- and middle-income regions and by quantifying international transfer schemes that can offset these burdens, thereby addressing an equity dimension often overlooked in broader analyses of trade and the environment.

We also contribute to the emerging literature on the distributional effects of climate change and carbon taxation. Krusell and Smith (2022) use an integrated assessment model to document large regional heterogeneity in climate damages, while Kotlikoff et al. (2024) study welfare implications of carbon taxes across regions and generations in an overlapping-generations framework. Closer to our approach, Calvacanti et al. (2024) and Finkelstein-Shapiro and Nuguer (2023) examine the sectoral and labor-market impacts of carbon taxation using closed-economy general equilibrium models calibrated to a small set of countries. We complement this literature by adopting a global perspective, covering the distributional effects of carbon policies across many countries and sectors.

Last, we also have points in contact with the integrated assessment model literature, which analyzes the aggregate effects of climate change and climate policy on the economy Nordhaus (1993, 2018), Golosov et al. (2014), Boyce (2018). We add an international economics dimension to the benefits and costs of climate policy.

⁷ We implement the three price floors proposed by the IMF: USD 75/tCO₂e for high-income economies, USD 50/tCO₂e for upper-middle-income economies, and USD 25/tCO₂e for lower-middle- and low-income economies. We consider the two IMF club structures (Canada, China, the EU, Great Britain, India, and the U.S.; and all G20 economies), a “key players” club (the EU, the U.S., China, India, Indonesia, Russia, Brazil, and South Africa), and a version in which all countries participate.

⁸ Other studies that analyze the effects of unilateral climate policy on trade include (Felder and Rutherford, 1993; Babiker, 2005; Elliott et al., 2010, 2013; Aichele and Felbermayr, 2012; Hémous, 2016; Böhringer et al., 2016; Kortum and Weisbach, 2021; Weisbach et al., 2023; Larch and Wanner, 2017, 2024).

2. Model

In this section, we develop a static, multi-country, multi-industry general equilibrium model of international trade in which governments have access to climate policy instruments. Trade and production follow an [Armington \(1969\)](#) structure with sectoral heterogeneity and input–output linkages, as in [Caliendo and Parro \(2015\)](#). The model incorporates fossil-fuel and agricultural emissions and features energy markets following [Garcia-Lembergman et al. \(2025\)](#) and [Farrokhi and Lashkaripour \(2025\)](#). Fossil-fuel extraction in each country relies on a fixed natural resource, generating decreasing returns to scale and an upward-sloping supply curve. We allow fossil-fuel prices to vary across countries, reflecting differences in local extraction costs, imperfect substitutability across origins, and segmented international markets, features consistent with observed dispersion in oil, coal, and natural-gas prices.

Our framework captures the near totality of greenhouse-gas emissions, those arising from energy use, agricultural production and fugitive emissions from fossil fuel extraction sectors, thus allowing consumers and firms to optimally adjust demand for final goods and inputs in response to climate policies.

There are $\mathcal{N}$ countries denoted by i (for origin) and j (for destination) producing a local-specific variety of S sectors denoted by s' (produced by origin i) and s (destination sector if in a value chain). Varieties of good s' produced by i and imported in country j are traded and subjected to iceberg trade costs $\tau_{is'j}$ such that $\tau_{is'i} = 1$.

Since production of fossil fuels differs from other sectors, we henceforth adopt the following notation:

$$S^M = \{\text{coal, oil, gas}\} \quad \text{and} \quad S \setminus S^M = \{\text{all other goods}\}.$$

Countries are endowed with an exogenous inelastic supply of labor L_i that is internationally immobile and, for each mining sector $s \in S^M$, a fixed natural-resource endowment M_{js} .

2.1. Carbon tax

A carbon tax is imposed on every good or service produced in the economy. The tax is rebated lump-sum to consumers either in the country of production or consumption, depending on the policy scenario.

We denote the ad-valorem equivalent carbon tax by $t_{is'j}^e$, where i is the origin country producing good or service s' for consumption in country j . The tax applies a monetary charge per ton of emitted CO₂-equivalent such that

$$\begin{aligned} t_{is'j}^e &= \frac{t_{is'j} e_{is'}}{p_{is'j}}, \\ e_{is'} &= \frac{\text{CO}_{2is'}}{q_{is'}}. \end{aligned} \tag{1}$$

where t denotes a dollar amount to be imposed on a ton of CO₂e (e.g. \$100), $p_{is'j}$ is the price of good s' produced in i and sold in country j and $e_{is'}$ denotes the emission intensity of production of good s' in country i , measured in tons of CO₂e per unit of output.

This formulation ensures that the carbon tax is equivalent to a per-unit tax on emissions. Indeed,

$$\begin{aligned} p_{is'j}(1 + t_{is'j}^e) q_{is'j} &= p_{is'j} q_{is'j} + t_{is'j} e_{is'} q_{is'j} = \\ &= p_{is'j} q_{is'j} + t_{is'j} \frac{\text{CO}_{2is'}}{q_{is'}} q_{is'j} = p_{is'j} q_{is'j} + t_{is'j} \text{CO}_{2is'}. \end{aligned}$$

where $q_{is'j}$ denotes the quantity of good s' shipped from origin i to destination j . Hence, total tax revenues equal the carbon price times total emissions.

2.2. Demand

Consumers in country j have Cobb–Douglas–CES preferences across sectors and sourcing origins, with utility reduced by unequal exposure to climate damages, as in [Shapiro \(2021\)](#):

$$\begin{aligned} U_j &= \left[\prod_{s'} U_{s'j}^{\beta_{s'j}} \right] \left[\frac{1}{1 + \delta_j (\text{CO}_2 - \text{CO}_2^B)} \right], \\ U_{s'j} &= \left(\sum_i a_{is'}^{1/\sigma_{s'}} q_{is'j}^{(\sigma_{s'}-1)/\sigma_{s'}} \right)^{\sigma_{s'}/(\sigma_{s'}-1)}. \end{aligned}$$

Here, $a_{is'}$ denotes an origin–sector–specific preference shifter, $\sigma_{s'}$ is the sectoral elasticity of substitution across origins, and $\beta_{s'j}$ is country j 's expenditure share on sector s' , satisfying $\sum_{s'} \beta_{s'j} = 1$. The parameter δ_j captures country j 's sensitivity to climate damages, while CO_2 and CO_2^B denote counterfactual and baseline emissions, respectively.

2.3. Production

Production is divided between mining sectors and non-mining sectors. Mining sectors - coal, oil, and gas - require a country-specific fixed natural resource and therefore display decreasing returns to scale. Non-mining sectors, in contrast, employ standard constant-returns-to-scale technology.

2.3.1. Non-mining goods

We assume perfect competition in the production of non-mining goods. Producing good s in country j requires labor and intermediate inputs from all sectors of the economy, combined through a Cobb–Douglas technology. Denoting by q_{js} the quantity of output of good s produced in country j , we have:

$$q_{js} = A_{js} \left(\frac{L_{js}}{\gamma_{j,LS}} \right)^{\gamma_{j,LS}} \prod_{s'} \left(\frac{m_{s'js}}{\gamma_{s'js}} \right)^{\gamma_{s'js}}$$

where L_{js} is the amount of labor used in j to produce s , $m_{s'js}$ the amount of intermediate input of sector s' used in j to produce s , and $\gamma_{s'js}$ the cost share of factor s' used in the production of s in country j , such that $\gamma_{j,LS} + \sum_{s'} \gamma_{s'js} = 1 \forall s$. A_{js} is an exogenous productivity shifter.

Intermediate inputs can be sourced domestically or imported, so that $m_{s'js}$ is a CES aggregate of country i -specific varieties of the input:

$$m_{s'js} = \left(\sum_i b_{is's}^{(1/\eta_{s's})} m_{is's}^{(\eta_{s's}-1)/\eta_{s's}} \right)^{\eta_{s's}/(\eta_{s's}-1)}$$

with $b_{is's}$ a preference shifter for variety i of product s' used in the production of s and $\eta_{s's}$ the elasticity of substitution between origins of good s' used in the production of good s .

2.3.2. Mining goods

Extraction of fossil fuels in sector $s \in S^M$ requires a fixed natural resource M_{js} . The production technology is given by:

$$q_{j,s} = (M_{js})^{\gamma_{j,Ms}} \left(\frac{L_{js}}{\gamma_{j,LS}} \right)^{\gamma_{j,LS}} \prod_{s'} \left(\frac{m_{s'js}}{\gamma_{s'js}} \right)^{\gamma_{s'js}},$$

where input use mirrors that of non-mining sectors and $\gamma_{j,Ms}$ is the natural resource share, with

$$\gamma_{j,Ms}, \gamma_{j,LS}, \gamma_{s'js} \in [0, 1], \quad \gamma_{j,Ms} + \gamma_{j,LS} + \sum_k \gamma_{s'js} = 1.$$

The presence of the fixed factor implies diminishing returns to variable inputs, leading to upward-sloping supply curves. A share $\gamma_{j,Ms}$ of total revenues in sector s of country j accrues as rents to natural-resource owners.

2.4. Equilibrium in levels

Utility maximization subject to the budget constraint implies that each firm faces standard CES demands:

$$q_{is'j} = a_{is'} [p_{is'} \tau_{is'j} (1 + t_{is'j}^e)]^{-\sigma_{s'}} I_{s'j} P_{s'j}^{\frac{c}{(\sigma_{s'}-1)}} \quad (2)$$

in which $p_{is'}$ is the price of a good from sector s' produced in country i , $I_{s'j} = \sum_i q_{is'j} p_{is'j} (1 + t_{is'j}^e)$ is country j 's sectoral expenditure on sector s' and $P_{s'j}^{\frac{c}{(\sigma_{s'}-1)}} = \left(\sum_i a_{is'} [p_{is'j} (1 + t_{is'j}^e)]^{(1-\sigma_{s'})} \right)^{\frac{1}{(1-\sigma_{s'})}}$ is a standard CES consumer price index.

Cost minimization subject to production technology constraints implies that the demand for intermediate goods is given by:

$$m_{is'js} = b_{is's} [p_{is'} \tau_{is'j} (1 + t_{is'j}^e)]^{-\eta_{s's}} E_{s'js} P_{s'js}^{\frac{\pi}{(\eta_{s's}-1)}} \quad (3)$$

in which $E_{s'js} = \sum_i m_{is'js} p_{is'j} (1 + t_{is'j}^e)$ is the expenditure of sector s firms in country j with sector s' inputs and $P_{s'js}^{\frac{\pi}{(\eta_{s's}-1)}} = \left(\sum_i b_{is's} [p_{is'j} (1 + t_{is'j}^e)]^{(1-\eta_{s's})} \right)^{\frac{1}{(1-\eta_{s's})}}$ is the appropriate index of intermediate goods.

For non-mining goods ($s \in S \setminus S^M$), ex-factory prices are determined by cost minimization and correspond to a Cobb–Douglas aggregation of the wage rate, w_i , and the prices of intermediate-input bundles:

$$p_{is'} = \frac{(w_i)^{\gamma_{i,LS'}}}{A_{is'}} \prod_k (P_{kis'})^{\gamma_{kis'}}, \quad s' \in S \setminus S^M \quad (4)$$

For mining goods ($s \in S^M$), solving the cost-minimization problem yields the following upward-sloping supply curve:

$$Y_{js} = p_{js} q_{js} = M_{js} p_{js}^{\frac{1}{\gamma_{j,Ms}}} w_j^{-\frac{\gamma_{j,LS}}{\gamma_{j,Ms}}} \prod_k (P_{kjs}^{\pi})^{-\frac{\gamma_{kjs}}{\gamma_{j,Ms}}}, \quad s \in S^M \quad (5)$$

Market clearing implies that the consumer balance equates total expenditure on final goods, I_j , to the sum of the wage bill, rents from natural-resource ownership, and revenues from carbon taxes collected on final and intermediate goods.

Under the scenario in which the tax is paid in the destination country (“consumer tax”), this condition writes as

$$I_j = \sum_s w_j L_{js} + \sum_s \gamma_{j,Ms} Y_{js} + \sum_{i,s'} p_{is'j} q_{is'j} t_{is'j}^e + \sum_{i,s',s} p_{is'j} m_{is'js} t_{is'j}^e + D_j \quad (6)$$

in which D_j is an exogenous and fixed deficit for each country, with $\sum_j D_j = 0$.

In the alternative case where the tax is levied in the origin country (“producer tax”), the corresponding balance condition becomes

$$I_j = \sum_s w_j L_{js} + \sum_s \gamma_{j,Ms} Y_{js} + \sum_{i,s'} p_{js't} q_{js'i} t_{js'i}^e + \sum_{i,s'} p_{js't} m_{js'is} t_{js'i}^e + D_j \quad (7)$$

Market clearing also implies that in the producer balance expenditure with intermediate inputs, wages and, in the case of the mining sector, rents to owners of the specific factor, $E_{js} = \sum_{s'} E_{s'js}$, is equal to the revenue obtained from domestic and foreign sales:

$$E_{js} = \sum_i \left(p_{jsi} \left(q_{jsi} + \sum_{s'} m_{jsis'} \right) \right) \quad (8)$$

By summing Eqs. (6) and (8) we obtain the country trade balance, where taxes cancel out because they are domestic:

$$\sum_{i,s'} p_{is't} q_{is'j} + \sum_{i,s',s} p_{is't} m_{is'js} = D_j + \sum_{i,s} \left(p_{jsi} \left(q_{jsi} + \sum_{s'} m_{jsis'} \right) \right) \quad (9)$$

Last, labor market clearing in each country implies that:

$$\sum_s \gamma_{j,LS} \frac{E_{js}}{w_j} = L_j \quad (10)$$

We define an equilibrium under carbon taxes $\{t_{is'j}\}$ as a set $\{q_{is'j}, m_{is'js}, p_{is}, I_j, E_{js}, w_j\}$ that satisfies equilibrium conditions Eqs. (2)–(6) (or Eq. (7)), Eqs. (8) and (10) for all i, j, s' and s .

2.5. Equilibrium in changes

In order to take the model to the data, we use Dekle et al. (2007)’s “exact hat algebra”, which is now standard in the literature. This involves re-writing variables as linear changes from the baseline. In what follows, a baseline version of a variable x is denoted by x^B . The proportional change is then given by $\tilde{x} = x/x^B$. Observe that this procedure has the advantage of eliminating the parameters that are hard to observe in the data, such as preference shifters $a_{is'}$ and $b_{is's}$, productivity shifters A_{js} and iceberg trade costs $\tau_{is'j}$, thus simplifying the quantitative analysis. It also implies that the model perfectly matches the global pattern of trade and production described by the data in the baseline.

Following this procedure, changes in the demand for final goods, the demand for inputs, price indexes, and ex-factory prices are given by:

$$\tilde{q}_{is'j} = \tilde{I}_{s'j} [\tilde{p}_{is'} (1 + t_{is'j}^e)]^{-\sigma_{s'}} \tilde{P}_{s'j}^{\frac{c}{(\sigma_{s'}-1)}} \quad (11)$$

$$\tilde{P}_{s'j}^{\frac{c}{(\sigma_{s'}-1)}} = \left(\sum_i [\tilde{p}_{is'} (1 + t_{is'j}^e)]^{(1-\sigma_{s'})} \left(\frac{q_{is'j}^B p_{is'js}^B}{I_{s'j}^B} \right) \right)^{\frac{1}{(1-\sigma_{s'})}} \quad (12)$$

$$\tilde{m}_{is'js} = \tilde{E}_{s'js} [\tilde{p}_{is'} (1 + t_{is'j}^e)]^{-\eta_{s'}} \tilde{P}_{s'js}^{\frac{\pi}{(\eta_{s'}-1)}} \quad (13)$$

$$\tilde{P}_{s'js}^{\frac{\pi}{(\eta_{s'}-1)}} = \left(\sum_i [\tilde{p}_{is'j} (1 + t_{is'j}^e)]^{(1-\eta_{s'})} \left(\frac{m_{is'js}^B p_{is'js}^B}{E_{s'js}^B} \right) \right)^{\frac{1}{(1-\eta_{s'})}} \quad (14)$$

$$\tilde{p}_{js} = \tilde{w}_j^{\gamma_{j,LS}} \prod_{s'} \tilde{P}_{s'js}^{\gamma_{s'js}}, \quad s \in S \setminus S^M \quad (15)$$

$$\tilde{Y}_{js} = \tilde{p}_{js} \tilde{q}_{js} = (\tilde{p}_{js})^{\frac{1}{\gamma_{j,Ms}}} \tilde{w}_j^{-\frac{\gamma_{j,LS}}{\gamma_{j,Ms}}} \prod_k \left(\tilde{P}_{kjs}^{\frac{\pi}{\gamma_{kjs}}} \right)^{-\frac{\gamma_{kjs}}{\gamma_{j,Ms}}}, \quad s \in S^M \quad (16)$$

Changes in the consumer balance equations, expressed in hat algebra, are given by the following relationships. Under the consumer-tax regime (tax paid at destination), we have

$$\tilde{I}_j = \sum_s \tilde{w}_j \tilde{L}_{js} (w_j^B L_{js}^B) + \sum_{i,s'} \tilde{p}_{is'} \tilde{q}_{is'j} t_{is'j}^e (p_{is'j}^B q_{is'j}^B) + \sum_{i,s'} \tilde{p}_{is'} \tilde{m}_{is'js} t_{is'j}^e (p_{is'j}^B m_{is'js}^B) + D_j^B \quad (17)$$

Under the producer-tax regime (tax paid at origin), the corresponding relationship is

$$\tilde{I}_j = \sum_s \tilde{w}_j \tilde{L}_{js} (w_j^B L_{js}^B) + \sum_{i,s'} \tilde{p}_{js'} \tilde{q}_{js'i} t_{js'i}^e (p_{js'i}^B q_{js'i}^B) + \sum_{i,s'} \tilde{p}_{js'} \tilde{m}_{js'is} t_{js'i}^e (p_{js'i}^B m_{js'is}^B) + D_j^B \quad (18)$$

Finally, changes in the producer balance equation are given by

$$\tilde{E}_{js} = \sum_i \left(\tilde{p}_{js} \tilde{q}_{jsi} (p_{jsi}^B q_{jsi}^B) + \sum_{s'} \tilde{p}_{js} \tilde{m}_{jsis'} (p_{jsi}^B m_{jsis'}^B) \right) \quad (19)$$

$$\sum_s \frac{\tilde{E}_{js}}{\tilde{w}_j} L_{js}^B = L_j \quad (20)$$

In [Appendix C](#), we show that the model expressed in changes can be reduced to a parsimonious $N \times S$ system that we use to explore the consequences of climate policies and international transfers that account for distributional concerns.

2.6. International transfers

A main goal of our analysis is to quantify international monetary transfers that could be used to balance the costs involved in the widespread adoption of climate policies. In this subsection, we explain how we formally introduce such transfers in the theoretical framework outlined above.

First, notice that domestic real income is given by total expenditure with final goods deflated by the domestic consumer price index. Hence, the associated real income changes are defined by:

$$\tilde{W}_j = \frac{\tilde{I}_j}{\Pi_{s'} \tilde{P}_{s'j}^{\beta_{s'j}}} \quad (21)$$

We use this measure of real income to solve for two types of contribution schemes. First, an “equal-cost” scenario, which makes sure that the adoption of a global carbon tax leads to uniform real income changes across countries irrespective of their present-day relative contribution to total GHG emissions. Second, a “historic polluter pays” transfer scheme that ensures that the real income costs of climate policies are proportional to the country’s cumulative GHG emissions since the onset of the Industrial Revolution. Next, we provide more details on how we formally model both types of schemes.

Equal Cost Transfers We define international transfers as an exogenous income source at the country level, T_j . In the “equal cost transfer” case, we compute transfers paid or received such that the real income changes induced by the carbon tax are equalized across countries. Formally, we want to solve the general equilibrium such that:

$$\tilde{W}_j = \tilde{W} \quad (R1)$$

$$\sum_j T_j = 0 \quad (R2)$$

Restriction (R1) implies that cross-country real income changes that result from the adoption of a global carbon tax are the same. Restriction (R2) is a zero-sum condition that makes sure that transfers are balanced. By rewriting R1 we can explicitly solve for each country’s equal cost transfer:

$$T_j = H_j \frac{\sum_j G_j}{\sum_j H_j} - G_j \quad (22)$$

where $H_j = \left(I_j^B \times \Pi_{s'} \tilde{P}_{s'j}^{\beta_{s'j}} \right)$ and $G_j = \sum_s w_j L_{sj} + \sum_s \gamma_{j,Ms} Y_{j,s} + \sum_{is'} p_{is'j} q_{is'j} t_{is'j}^e + \sum_{is's} p_{is'j} m_{is'js} t_{is'j}^e + D_j$. To back out T_j in this case, we add Eq. (22) as a fixed point to the algorithm described in [Appendix C](#) together with condition (R2).

Historic polluter-pays transfer For this type of contribution scheme, we want to compute country-level international transfers T_j that are proportional to historical emissions of GHG in each country. We achieve this by solving the general equilibrium such that:

$$(\tilde{W}_j - 1) W_j^B = \Gamma \left[\frac{Z_j}{\sum_j Z_j} \right] \quad (R3)$$

$$\sum_j T_j = 0 \quad (R4)$$

where W_j^B is a measure of real income in the baseline, Z_j are historical greenhouse gas emissions in country j and Γ is an endogenous proportionality factor. Restriction (R3) implies that changes in real income are proportional to the share of historical emissions in each country. Restriction (R4) is a zero-sum condition as before. Again, we can use the two conditions to explicitly solve for transfers T_j :

$$T_j = H_j - G_j + \Omega_j^B H_j \frac{\sum_j (G_j - H_j)}{\sum_j \Omega_j^B H_j} \quad (23)$$

where G_j and H_j are defined as before and $\Omega_j^B = \left[\frac{Z_j}{\sum_j Z_j} \right] \frac{1}{c_j^B}$. Notice that the zero-sum condition determines the proportionality factor:

$$\Gamma = \frac{\sum_j (G_j - H_j)}{\sum_j \Omega_j^B H_j} \quad (24)$$

To back out transfers in this case, we add expressions Eqs. (23) and (24) as a fixed point to the algorithm described in Appendix C.⁹

2.7. Carbon emissions accounting

We account for greenhouse-gas emissions arising directly from the production of fossil fuels (coal, oil, and natural gas) and agricultural goods. Our measure captures emissions from extraction and consumption of fossil fuels and agricultural production (Scope 1).¹⁰ Downstream sectors generate emissions only indirectly through their use of these carbon-intensive inputs. For example, the air transport sector emits in proportion to the fossil fuels embodied in the refined petroleum it purchases: crude oil is extracted in the oil sector, refined into kerosene in the coke–petroleum sector, and then used by air transport. Only the flow of oil is taxed. Depending on the taxation scheme, either the country of consumption or the country of production obtains the tax revenue. The air transport sector itself does not pay carbon taxes in either case.

Total emissions CO_{2j} are given by:

$$CO_{2j} = CO_{2j}^{fossil} + CO_{2j}^{agric}$$

where

$$CO_{2j}^{fossil} = \sum_{s' \in SM} \frac{e_{js'}^{fossil}}{p_{js'}} Y_{js'}$$

$$CO_{2j}^{agric} = \frac{e_j^{agric}}{p_{js'}} Y_{js'}, \quad s' = agriculture$$

Here, $e_{js'}^{fossil}$ denotes tons of CO_2e emitted per unit of fossil-fuel output in sector s' of country j , and e_j^{agric} the CO_2e emitted per unit of agricultural production.

This formulation aligns the carbon tax with actual emissions at the point of extraction, ensuring that our policy experiments tax physical emissions generated within each sector and country, while embodied upstream or downstream emissions are not included.

During our analysis, we also refer to measures of total emissions within the supply chain. Formally, following the notation in our theoretical framework, total emissions intensity along the supply chain for a good produced in country i and sector s' , denoted $TE_{is'}$, is defined as the unique positive solution to:

$$p_{is'} q_{is'} TE_{is'} = p_{is'} q_{is'} \tilde{e}_{is'} + \sum_{js} m_{jsis'} p_{jsi} TE_{js} \quad (25)$$

where $\tilde{e}_{is'}$ is emission intensity per dollar of output.¹¹ This equation decomposes total emissions into direct emissions at the final production stage and the cumulative emissions embodied in all intermediate inputs.

3. Data

This section describes the data sources, the data treatment, and the calibration of the parameters.

3.1. Data sources

Data on production, consumption, and trade flows come from the OECD Inter-Country Input-Output (ICIO) tables (OECD, 2023). The ICIO tables report data on intermediate input use, final demand, consumption expenditure of various accounting units, value added and gross output in 45 distinct sectors for 67 countries, including a Rest of the World aggregate, from 1995 to 2018. As the ICIO tables only have one fossil fuel sector (mining, energy, code 05T06), we use the Exiobase input–output tables to disaggregate the ICIO fossil fuel sector into oil, gas, and coal (Stadler et al., 2025). This allows us to keep the more detailed country list of the ICIO data while adding detail on the fossil fuel sectors. The model is calibrated with trade, production and emissions data in 2018. We use the remaining years to back out some key parameters of the model, as we explain below.

For the data on greenhouse gas emissions in CO_2 equivalents for the mining sectors, we follow Garcia-Lembergman et al. (2025) and use the values from the U.S. Energy Information Administration, which reports $e_{coal} = 3.81$, $e_{gas} = 2.06$, and $e_{oil} = 2.94$. We supplement this data to include fugitive emissions with the Emissions Database for Global Atmospheric Research (EDGAR) (European Commission, 2023). The non-fuel-related emissions for the agricultural sector stem from FAOSTAT Domain Emission Totals (FAO, 2023b). We describe how we combine the datasets in detail in the Appendix under Appendix A. Appendix B in the Appendix provides a description of changes in emissions by category (fuel, fugitive emissions, agriculture) between 1995 and 2018.

⁹ Since we cannot observe sector-level price indexes in the data, we use consumption data in the baseline to solve for transfers in the “historical polluter pays” scheme. As shown in Fig. 20, this procedure leads as intended to a counterfactual equilibrium in which carbon taxes and transfers imply a perfect correlation between changes in real income and the historical cross-country pattern of GHG emissions.

¹⁰ See Garcia-Lembergman et al. (2025) for a discussion of alternative accounting methods, such as consumption-based emissions.

¹¹ We express emission intensities per dollar of output in this case because physical quantity data is readily available only for agriculture and fossil fuels.

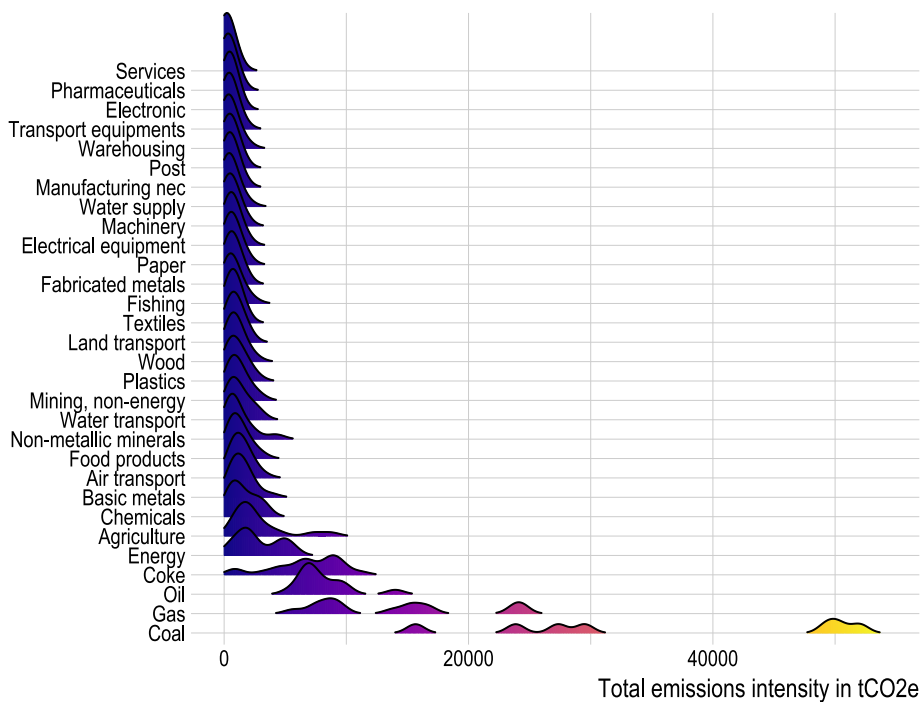


Fig. 1. Total emissions intensity across sectors.

Note: This figure shows the emissions of the total supply chain for each sector in tCO₂ per mn USD. Country-sector pairs with a sectoral output share below 1% are omitted in the figure for readability.

We obtain price data for agricultural goods and fossil fuels from multiple sources. Agricultural prices are constructed using data on producer prices, quantities, and exchange rates from [FAO \(2023a\)](#), from which we compute an agricultural price index for each country. Fossil-fuel prices are derived from a harmonized dataset that we assemble using the *IEA World Energy Balances Highlights* (2024). For 2018, we collect production volumes for coal, crude oil, and natural gas, convert all quantities into tonnes of oil equivalent, and merge them with the corresponding monetary output values to obtain implied country-specific prices for each fuel. This procedure yields a consistent country–fuel price matrix used in the model. Further details are provided in [Appendix A](#).

[Fig. 1](#) shows the total emissions intensities by sector. The *y*-axis shows all sectors in the final data set, where we group all the services sectors for readability. The *x*-axis shows the total emissions intensity of a sector in tons of CO₂e across countries. While emissions originate in the mining (oil, coal, gas) and agricultural sectors, the total emissions intensity shows the emissions intensity of the whole supply chain. For the air transport sector, this includes the emissions from the oil sector and the emissions occurring when refining oil in the coke, petroleum sector. There are two main takeaways. First, there is large variation in total emission intensities across sectors. The mining and energy sectors have the largest emissions intensities, followed by agriculture and chemicals. Second, there is a large variation in total emission intensities across different countries. Notice that this variation is more pronounced for highly carbon intensive sectors, as energy and agriculture.¹²

3.2. Calibration

We calibrate the model using data for the year 2018. The following groups of parameters require calibration: substitution elasticities, household expenditure shares, production function parameters, emission intensities and climate damage parameters.

Elasticities of substitution. We assume a common elasticity of substitution across origins for consumption and intermediate inputs, such that $\sigma_{s'} = \eta_{s's} = \eta_{s'}$. Our baseline estimates follow the procedure in [Caliendo and Parro \(2015\)](#). To assess robustness, we consider two additional specifications: (i) estimation using the approach in [Fontagné et al. \(2022\)](#), and (ii) direct use of the elasticities reported in [Fontagné et al. \(2022\)](#), mapped to our industry classification. [Table D.17](#) in the Appendix summarizes the results. Across all approaches, estimated elasticities fall within the range commonly found in the trade literature.

Household expenditure shares. The Cobb–Douglas utility parameters $\beta_{s'j}$ are calibrated directly from expenditure data. For each country *j* and sector *s'*, we compute $\beta_{s'j}$ as the share of consumption expenditure in that sector relative to total absorption.

¹² Plotting the density of the emissions intensity in % deviation from the sectoral mean in [Fig. 17](#) in the Appendix, we can observe a positive skew. That means that there are a few countries that have a much higher emission intensity than the average country in a given sector.

Production function parameters. We calibrate the Cobb–Douglas production parameters using input–output data. Intermediate input shares $\gamma_{s'js}$ are given by the ratio of sector- s' intermediate expenditures to gross output in sector s and country j . For the mining sectors, the natural-resource share $\gamma_{j,Ms}$ is calibrated following Garcia-Lembergman et al. (2025). Consistent with their approach, we assume a common natural-resource share across countries and adopt their estimates of the global supply elasticity for coal, natural gas, and crude oil, a parameter that is isomorphic to the share of natural resources in mining. This yields $\gamma_{j,Ms} = 0.25$, $s \in S^M$. Labor shares $\gamma_{j,Ls}$ are calibrated using value added data for non-mining sectors, while for mining sectors they are obtained as the residual ensuring that all Cobb–Douglas shares sum to one.

Emission intensities. We compute country-sector-level emission intensities for the fugitive emissions of fossil fuels and agricultural production emissions by dividing direct emissions by the corresponding quantity produced. Production quantities are obtained by deflating nominal output with sectoral prices. Formally,

$$e_{is'} = \frac{CO_{2is'}}{q_{is'}} = \frac{CO_{2is'}}{Y_{is'}/p_{is'}}, \quad (26)$$

where $e_{is'}$ denotes the direct emission intensity of sector s' in country i , $CO_{2is'}$ is total emissions from that sector, $Y_{is'}$ is nominal output, and $p_{is'}$ is the sectoral price. Section 3.1 provides details on all data sources used to construct these variables. The fugitive emissions intensities for the fossil fuel sector are added to the emissions intensities from the U.S. Energy Information Administration as discussed in Section 3.1.

Country-specific climate damage parameters. We obtain data for country-specific climate damages from Ricke et al. (2018). This data set provides heterogeneous social cost of carbon (scc) estimates for different countries in a plethora of scenarios.¹³ In line with the carbon tax in the main part of the paper, we set the global social cost of carbon to 100 USD/tCO₂e. We then calculate $\delta_j = \frac{scc_{sccj}}{\sum_j scc_j GDP_j}$, thus translating the country-specific social cost of carbon into multiplicative changes in GDP and normalizing the change at global scale to our chosen scc .

4. Results

4.1. Carbon taxes

We begin by presenting the results for the counterfactual scenario of a 100 USD/tCO₂e carbon tax, focusing first on its macroeconomic impacts. Overall, the carbon tax is highly effective at reducing emissions while imposing relatively modest economic costs. In our benchmark policy, global emissions decrease by 52%,¹⁴ real incomes decline by 0.6%, and global gross output remains broadly unchanged (increasing by 0.1%). Notably, the share of output traded internationally remains constant at 13%, underscoring the central role of international trade in mediating the adjustment to carbon pricing. Firms and consumers rely on access to foreign varieties and production networks to reallocate activity toward cleaner sources.¹⁵

4.1.1. Country effects

The macroeconomic effects, however, conceal significant heterogeneity across countries. Fig. 2 plots real income changes against GDP per capita, with panel (a) reporting results for the consumer-tax scenario (tax rebated to destination) and panel (b) for the producer-tax scenario (tax rebated to origin).

Under the consumer tax, most low- and middle-income countries experience sizable real income losses, whereas many high-income countries see gains or comparatively smaller declines. This pattern reflects the fact that lower-income economies tend to be more specialized in upstream, energy-intensive, and agricultural production, activities that become relatively more expensive with the carbon tax.

In the producer-tax scenario, low- and middle-income countries still experience larger real income losses on average, but the gains previously observed for many high-income countries are now far less pronounced, and most of them instead face modest declines. A notable difference in this regime is the large real income changes for several fossil-fuel exporters, most prominently Brunei, Saudi Arabia, and Norway. Because carbon taxes are collected at the origin, these countries retain substantial tax revenue that partly offsets the contraction in fossil-fuel production. In addition, oil exporters with relatively low emission intensities benefit from a reallocation of global demand toward cleaner fossil-fuel sources, further mitigating their losses or even generating positive income effects.

To fully assess the impact of carbon taxes, one also needs to account for how they mitigate climate damages. Fig. 3 illustrates the trade-off between the cost of climate action and the cost of inaction for both tax regimes. The y-axis represents the real income

¹³ We use the following parameters and scenarios: ssp=SSP3, rcp=RCP85, run=BHM-LR, prtp=2, dmfuncpar=estimates, climate=expected. For a detailed explanation of these options see the readme file from Ricke et al. (2018).

¹⁴ For comparison, Garcia-Lembergman et al. (2025) find that a 100 USD/tCO₂e carbon tax reduces global emissions by 45.6% and real income by 0.6%. Although our framework is closely related, several modeling differences help explain the variation in results. In particular, we allow for cross-country variation in fossil-fuel prices, incorporate emissions from agricultural production, account for intermediate-input use in fossil-fuel extraction, and evaluate the policy in a different reference year.

¹⁵ As shown in Le Moigne et al. (2024), within a similar framework, roughly one-third of the emission reductions induced by a carbon tax arise through the environmental gains from trade, as production shifts toward countries with stronger environmental comparative advantage.

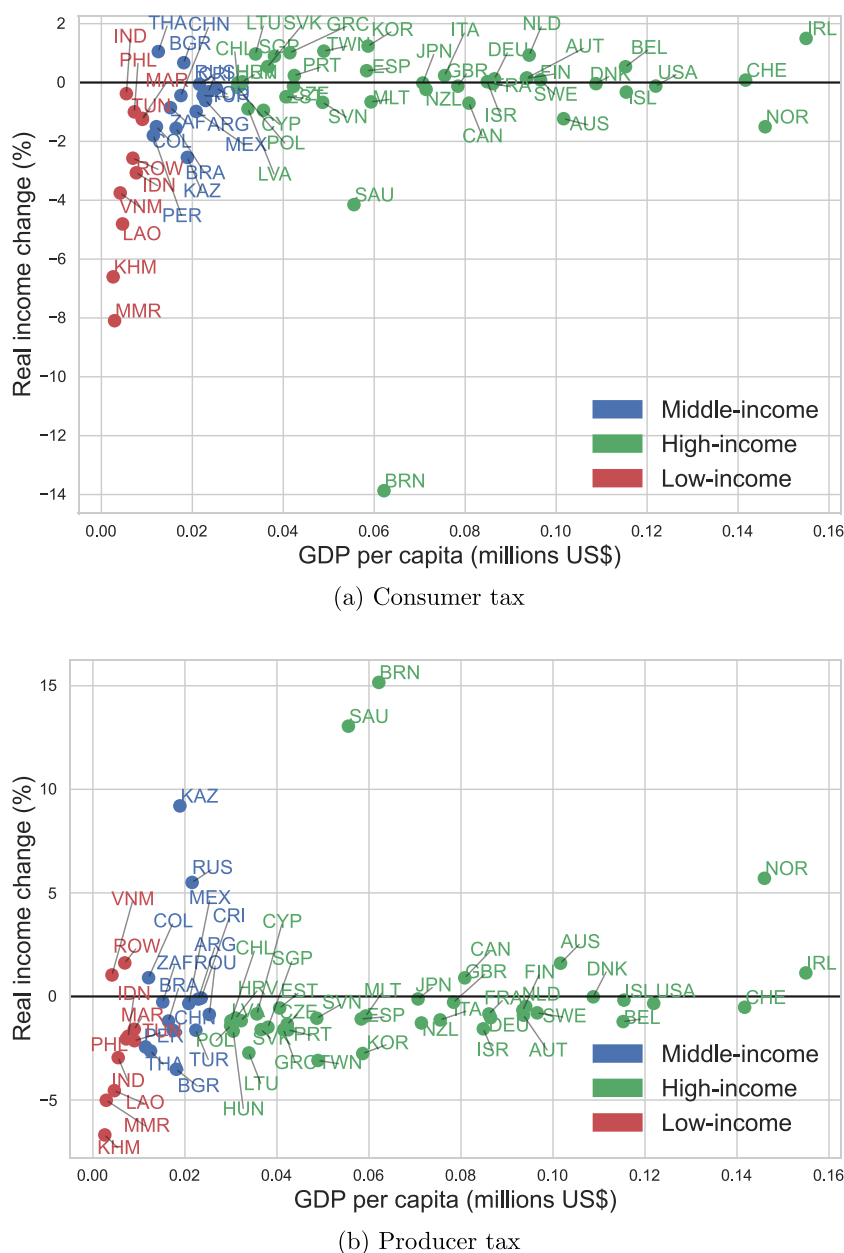
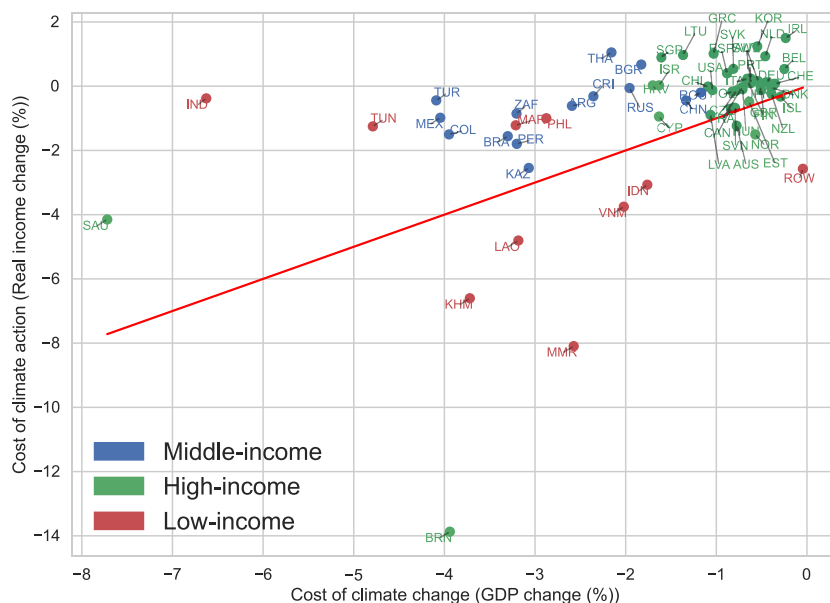


Fig. 2. Real income changes under consumer and producer carbon taxes.

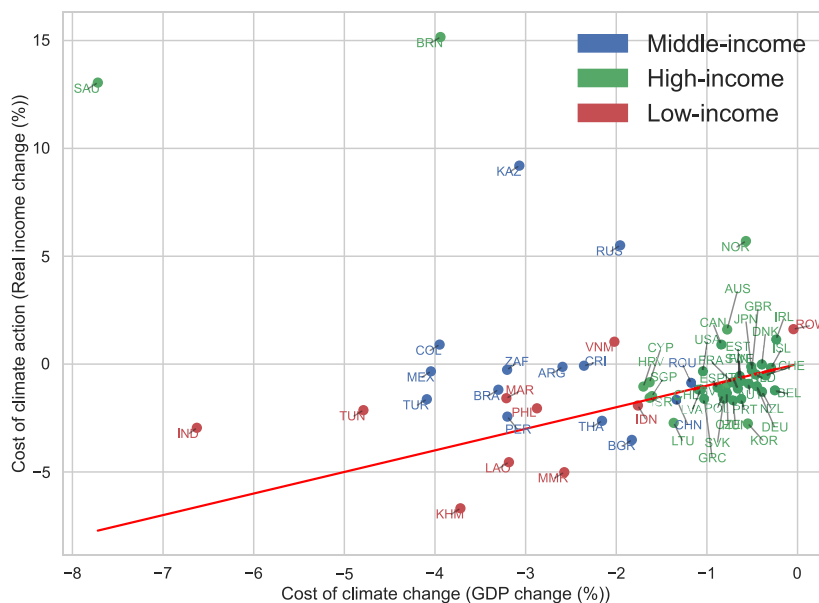
Note: This figure compares changes in real income across countries following the implementation of carbon taxes, categorized by GDP per capita levels. Panel (a) reports results under a consumer tax (destination-based) and panel (b) under a producer tax (origin-based).

change resulting from a 100 USD/tCO₂e carbon tax (the cost of climate action), while the x-axis displays the cost of climate change as a percentage of GDP. Two outliers, Taiwan (31% climate damage) and Malta (14% climate damage), are excluded for clarity. Countries positioned above and to the left of the red line face greater costs from climate damages than from climate action, while those below and to the right face higher costs from climate action. Fig. 4 extends this analysis by presenting the welfare changes from a 100 USD/tCO₂e carbon tax, accounting for both real income changes and the avoided costs of climate damages.

Results show that carbon taxes generate average welfare gains of 1.8% in the consumer-tax regime and 2.3% in the producer-tax case, with the latter displaying greater cross-country dispersion. Incorporating climate damages into the welfare metric makes climate policy more attractive for low- and middle-income countries, as these economies are expected to bear disproportionately large climate-related losses in the absence of mitigation. Nevertheless, this adjustment is not sufficient to ensure welfare gains for all low-income countries, underscoring that climate action still imposes meaningful burdens on many of the most vulnerable economies.



(a) Consumer tax



(b) Producer tax

Fig. 3. Cost of climate action vs. cost of climate change.

Note: This figure compares the cost of climate action with the cost of climate inaction across countries under a 100 USD/tCO₂e carbon tax. The y-axis reports the real income change from the carbon tax (cost of climate action), while the x-axis shows the projected cost of climate damages as a percentage of GDP. Two outliers – Taiwan (31% climate damage) and Malta (14% climate damage) – are excluded for clarity. Panel (a) corresponds to the consumer-tax regime and panel (b) to the producer-tax regime.

The differences between the two tax regimes illustrate that the point of taxation continues to matter even when climate damages are taken into account. Interestingly, the divergence is most pronounced among high-income countries: many experience welfare losses under the producer-tax regime. Although these countries tend to have relatively low emission intensities – making mitigation itself less costly – failing to retain carbon-tax revenue raises the overall cost of climate action for them.

A key reason why many low- and middle-income countries bear large real income and total welfare losses is that they face substantially higher effective carbon tax rates. Fig. 5 reports the effective tax rate, calculated as the total tax paid by producers and consumers in a country divided by the country's total expenditure, with panels (a) and (b) corresponding to the consumer- and producer-tax regimes. The data show that effective tax rates in high-income countries typically range between 1%–2%, while in low-income countries, they can reach up to 7%. This reflects their greater reliance on energy-intensive production and consumption bundles, which exposes them more directly to the carbon tax. Importantly, effective tax rates look similar across the two regimes: changing the point of taxation alters where revenue is collected, but not the underlying emissions base that determines the aggregate tax burden.

We further investigate the impact of the carbon tax by examining how economic activity is reallocated across countries. Fig. 6 displays the net change in output together with the share of output that is reallocated across origins. For each country, the sum of the two bars represents the total gross change in output: the solid bar shows the net change, while the hatched orange bar captures the reallocated component—that is, the share of output that shifts to or from foreign producers.¹⁶

Low- and middle-income countries exhibit both the largest net output declines and the strongest reallocation pressures. Most of them experience substantial gross output losses, whereas high-income economies typically display small net changes or modest gains. These large reallocation effects reflect the fact that the production structures of many low- and middle-income countries are less aligned with environmental comparative advantage (Le Moigne et al., 2024). As a result, the carbon tax requires more extensive adjustments in these economies—not only through real income losses, but also through significant shifts in the geographic and sectoral distribution of production.

We next document the impact of carbon taxes on real output changes for the different country-pair sectors. Fig. 7 plots the relationship between a sector's carbon intensity of production (horizontal axis, measured in tons of CO_2e per million USD of output) and the corresponding percentage change in real output under the carbon tax (vertical axis). Each point represents a country-sector pair, grouped into major sectoral categories: agro-food, raw materials, coal, manufactures, oil, and gas. Panel (a) displays the consumer tax case and panel (b), the producer tax case.

As expected, the results reveal a clear negative association: sectors with higher carbon intensity tend to experience larger contractions in real output. Fossil-fuel sectors (coal, crude oil, natural gas) exhibit the steepest declines, with particularly large output losses for coal in China and natural gas in Russia and Norway. Oil-exporting countries such as Saudi Arabia, Norway, and the United States also show substantial declines in their crude-oil sectors. By contrast, low-carbon sectors, such as manufactures and most agro-food activities, display relatively small output adjustments, reflecting their lower exposure to carbon pricing.

Changes in global trade are driven not only by shifts in consumption patterns, but also by adjustments firms make to reduce the emission intensity of their production, primarily through building cleaner supply chains. Fig. 8 highlights these firm-level adjustments by reporting the average changes in total emission intensities across sectors along the supply chain following the implementation of a 100 USD/t CO_2e carbon tax. All countries reduce their carbon intensity by at least 40%, highlighting the broad effectiveness of the policy. Among major economies, China exhibits particularly sharp reductions, reflecting the greater carbon intensity of its production structure and thus the stronger incentives to decarbonize. Because these adjustments are driven by tax incidence rather than tax collection, the patterns under the consumer- and producer-tax regimes are highly correlated.

4.2. Transfers

In this section, we explore international transfer schemes designed to compensate countries that incur losses from climate action. As shown in the previous section, developing countries disproportionately bear the economic burden of climate change mitigation policies. We demonstrate that moderate monetary transfers between countries can effectively redress this imbalance, ensuring a more equitable distribution of costs.

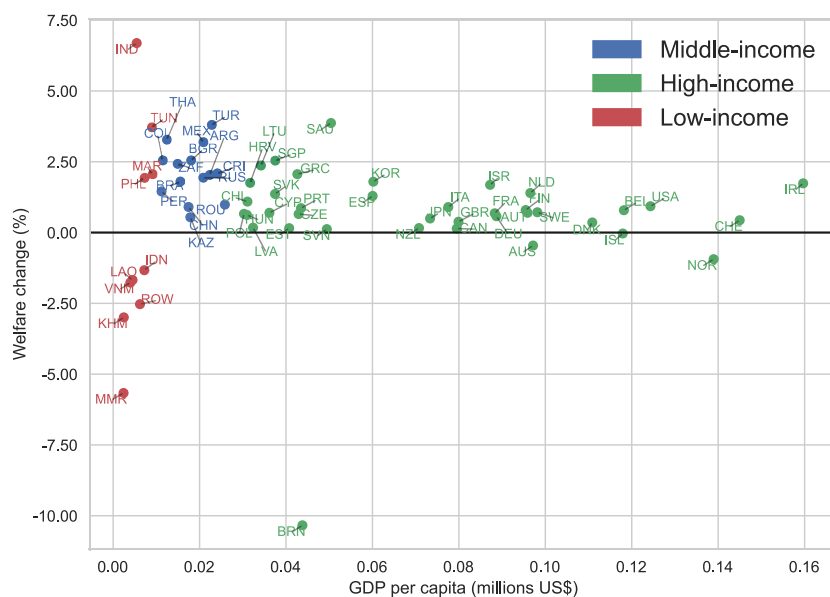
4.2.1. Equal cost transfers

Our first scenario is an “equal-cost” type of transfer, in which the goal is to perfectly equalize the real income costs of climate action. While this scenario does not fully implement the CBDR principle, as it does not account for differentiated responsibilities, it does ensure that at least low and middle-income countries do not bear the bulk of the economic costs.

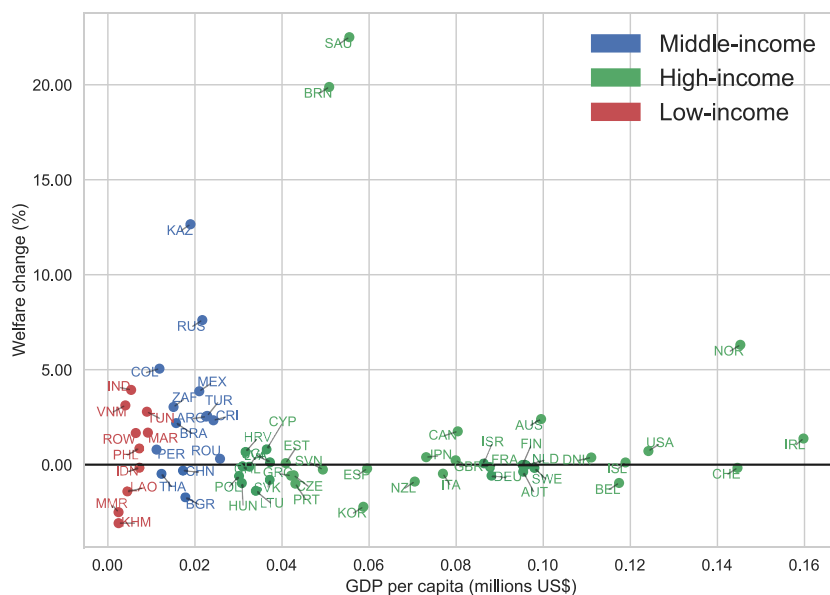
The results indicate that pairing the global common carbon tax with monetary transfers significantly smooths its overall impact compared to a scenario without transfers. Reduction in global emissions remains the same and the aggregate real income cost is identical. However, real income changes are uniformly negative but remain below 1% for all countries, ensuring that the economic burden is distributed equally, regardless of each country's relative contribution to the climate crisis.

Fig. 9 displays the per-capita transfers consistent with this equal-cost allocation under both the consumer- and producer-tax regimes. Negative values correspond to net contributions, while positive values reflect net receipts. Under the consumer tax, and with a few exceptions, high-income countries are required to transfer resources to low- and middle-income economies to equalize the economic burden of the tax. The largest contributors are the United States and European countries, while major beneficiaries include Kazakhstan, Russia, South Africa, and several Southeast Asian economies. A set of outliers, principally Brunei, Saudi Arabia, and Norway, also emerge as large recipients despite being high-income. This occurs because these fossil-fuel exporters face particularly

¹⁶ Intuitively, if bars represented firms, the solid portion would measure the net number of firm entries and exits in a country, while the hatched portion would represent firms that close domestically and reopen abroad (or vice versa). Their sum reflects the gross turnover in production locations.



(a) Consumer tax



(b) Producer tax

Fig. 4. Total welfare change.

Note: This figure reports total welfare changes from a 100 USD/tCO₂e carbon tax, combining real income effects with the avoided costs of climate damages. Panel (a) presents results under the consumer-tax regime (destination-based), and panel (b) under the producer-tax regime (origin-based).

large losses under the consumer-tax regime since the carbon price directly penalizes their core export activity. Since the equal-cost rule ignores geopolitical considerations and equalizes only economic incidence, these countries qualify as recipients.

Under the producer-tax regime, the pattern of transfers shifts substantially. Fossil-fuel producers become the main contributors, compensating much of the rest of the world including many high-income economies. The mechanism is straightforward: because the tax is collected at the point of production, fossil-fuel exporters retain large amounts of carbon-tax revenue and therefore experience sizable welfare gains; absent transfers, many high-income countries would face net losses and thus require compensation. Some high-income economies with relatively clean production structures, such as Switzerland and Japan, continue to be net contributors,

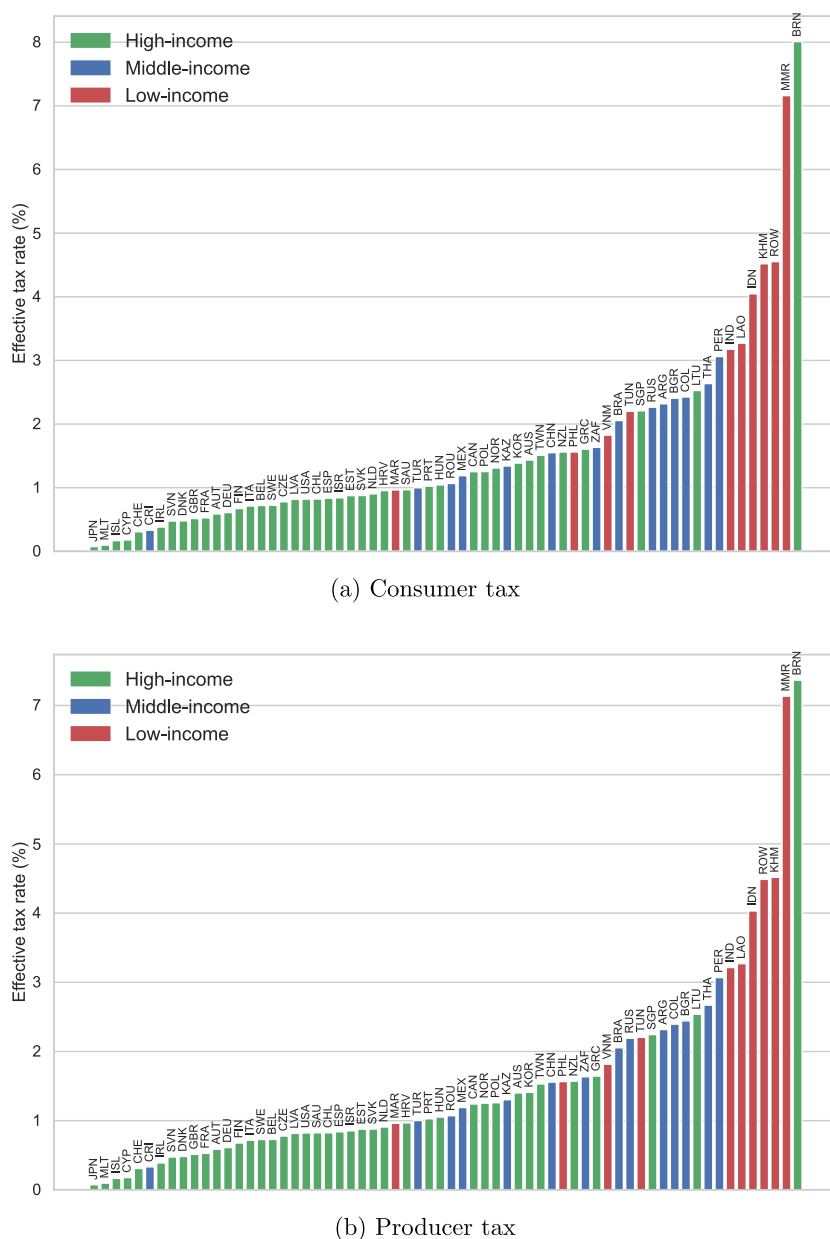


Fig. 5. Effective carbon tax rates under consumer and producer tax regimes.

Note: This figure displays the effective carbon tax rates across countries, grouped by income levels. The effective rate is calculated as the total tax burden – under each respective regime – divided by total national expenditure. Panel (a) corresponds to the consumer-tax case (destination-based) and panel (b) to the producer-tax case (origin-based).

though their per-capita payments are smaller than under the destination-based regime. Others, including Germany, Italy, and South Korea, switch signs and become net recipients. Low- and middle-income countries remain net beneficiaries overall, reflecting their higher effective tax burdens and structural exposure, although a few fossil-fuel exporters such as Kazakhstan and Russia remain exceptions and emerge as contributors.

To put our results in perspective, it is useful to compare the implied transfers with real-world mitigation commitments made by high-income countries. At the 15th Conference of Parties (COP15) in Copenhagen in 2009, high-income countries pledged to mobilize USD 100 billion per year by 2020 to support climate action in low- and middle-income regions. According to OECD monitoring, actual mobilized climate finance reached USD 83.3 billion in 2020—falling short of the pledge (OECD, 2022).

In contrast, our model implies substantially larger amounts would be required to equalize the economic costs of climate action under a global carbon tax. The transfers consistent with an equal-cost allocation amount to USD 231 billion under the consumer-tax

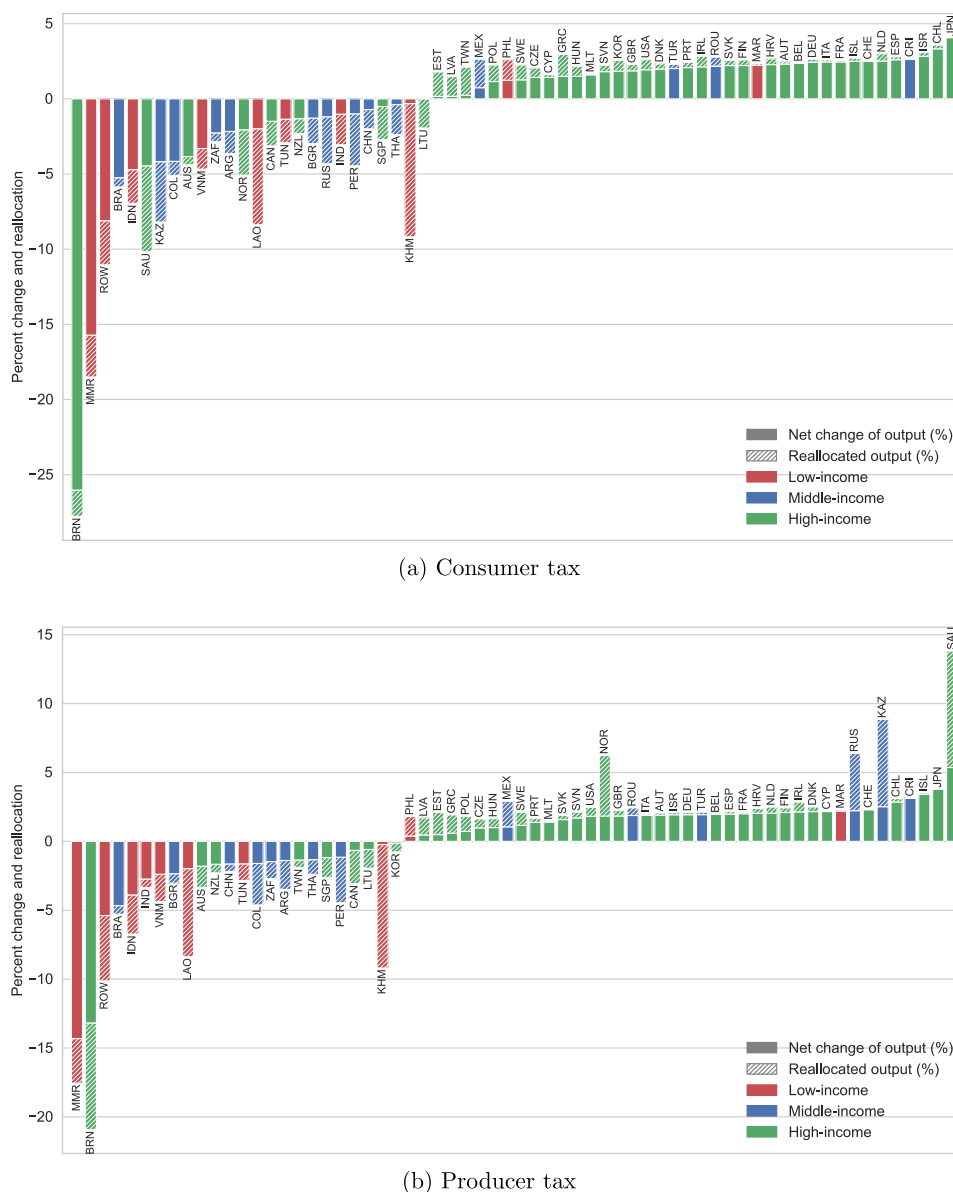


Fig. 6. Within-country output reallocation under consumer and producer carbon taxes.

Note: This figure illustrates the economic impact of carbon taxes on different countries, grouped by income levels. It shows the net change in output (solid bars) and the reallocation of output across sectors within each country (hatched bars). The sum of the two bars represents the total output change across all sectors. Panel (a) reports results for the consumer-tax case (destination-based) and panel (b) for the producer-tax case (origin-based).

regime and USD 391 billion under the producer-tax regime. Of these totals, flows from the Global North to the Global South would represent more than half of all transfers in the consumer-tax case (USD 165 billion), and roughly one-third in the producer-tax case (USD 107 billion). These magnitudes underscore the scale of redistribution needed to make globally efficient climate action politically acceptable across income groups.

While these figures exceed the current USD 100 billion pledge by a factor of three, the implied per-capita contributions for high-income countries remain within a range that is economically feasible. In the consumer-tax case, the largest per-capita payments arise in Ireland (about USD 2500 per person per year) and the Netherlands (around USD 1400). For other EU member states and the United States, annual contributions remain below USD 1000 per capita. On the receiving side, per-capita transfers are smaller, both because low- and middle-income countries tend to have larger populations and because equal-cost allocations distribute support across many recipients. For instance, Kazakhstan would receive roughly USD 600 per person, while Laos and Indonesia would receive less than USD 200.

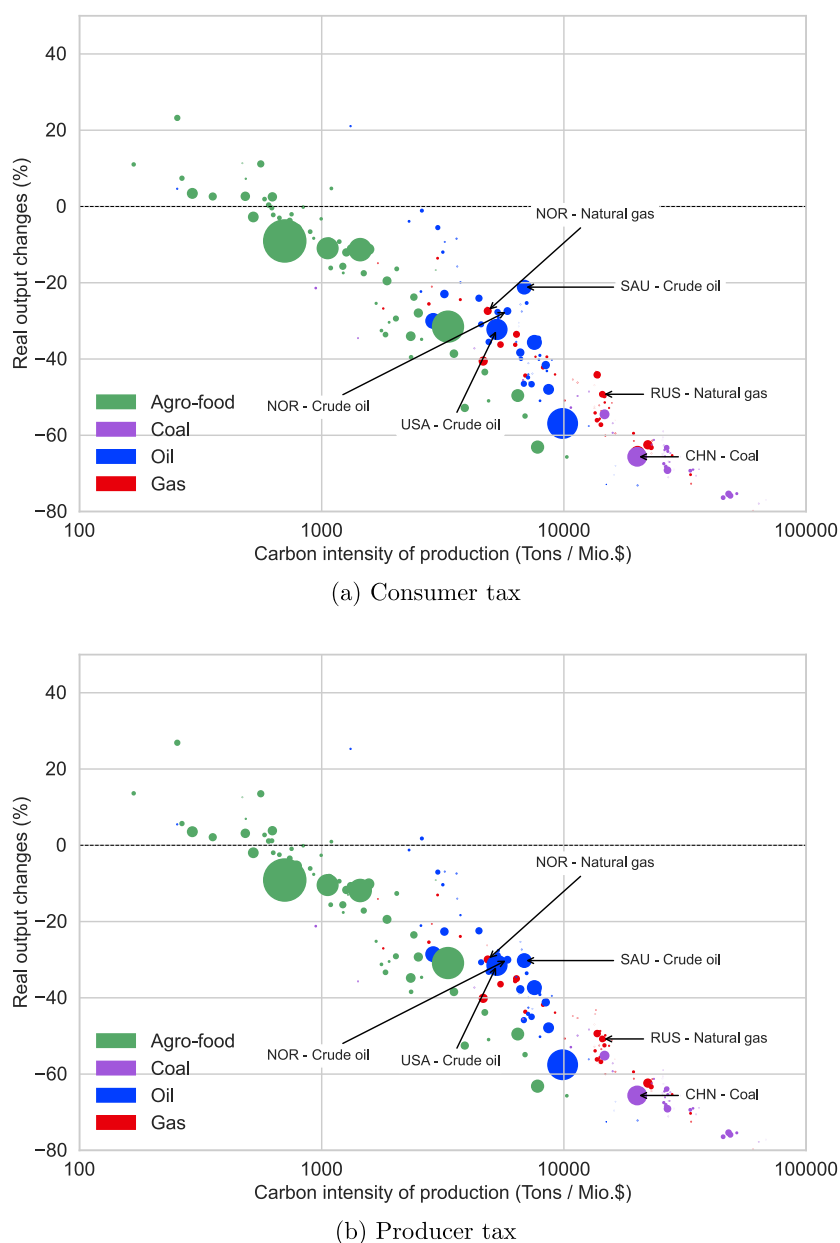
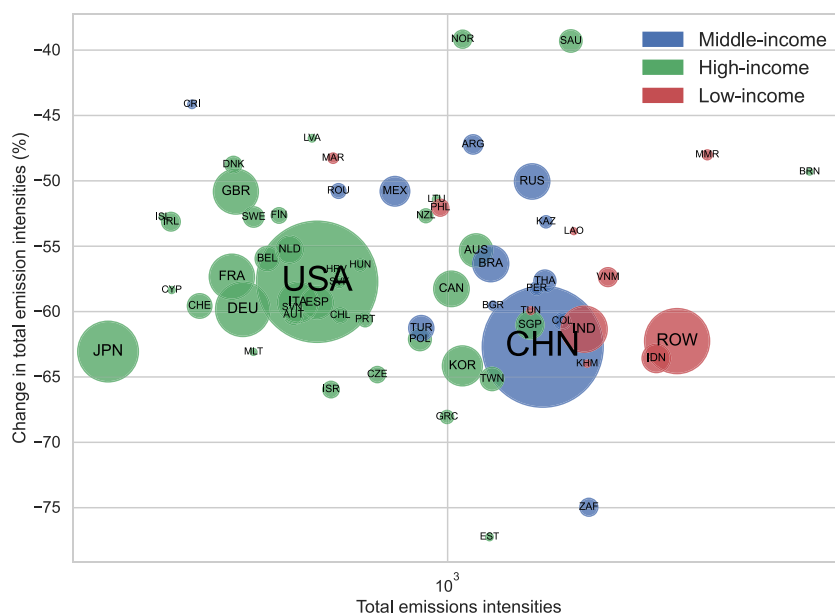


Fig. 7. Carbon intensity and real output changes.

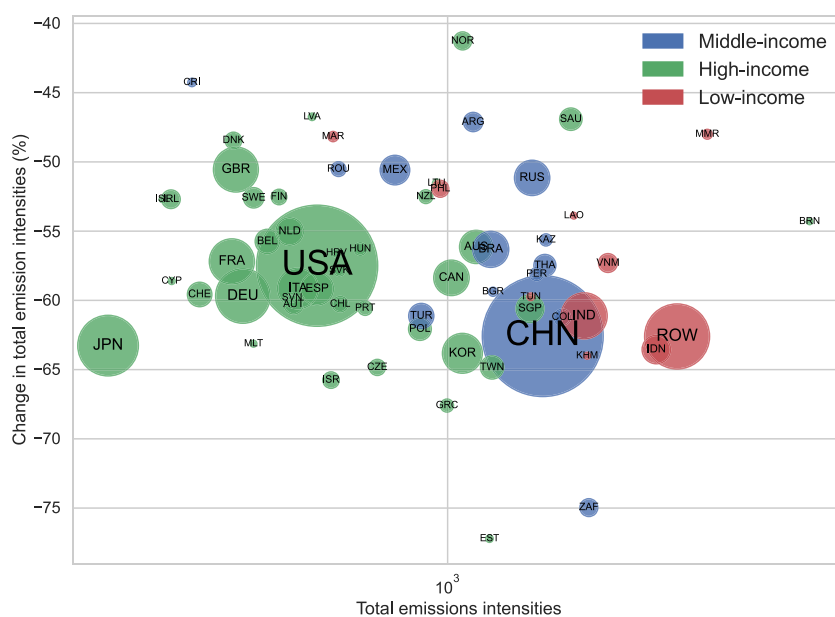
Note: This figure plots the percentage change in real output for each country-sector pair against the sector's carbon intensity of production (measured in tons of CO₂ per million USD of output). Points are grouped by broad sectoral categories, including agro-food, raw materials, manufactures, coal, oil, and natural gas. Selected country-sector combinations are labeled for reference.

In the producer-tax case, fossil-fuel producers face much larger per-capita contributions, reflecting the substantial welfare gains they obtain from retaining producer-tax revenue domestically. Norway and Brunei would contribute the most, with transfers approaching USD 8000 per person. In contrast, the required contributions for most other countries, including high-income economies not specialized in fossil-fuel extraction, remain broadly comparable to those in the consumer-tax regime, typically below USD 1000 per capita.

Overall, the equal-cost scenario demonstrates that, even without incorporating historical responsibility, distributing the economic burden of global climate action in a way that equalizes its real-income effects is attainable with per-capita contributions that are well within the fiscal capacity of high-income economies. That said, implementing such transfers in practice would require overcoming significant political, institutional, and geopolitical constraints, challenges that go beyond the scope of our quantitative exercise but are central to real-world climate negotiations.



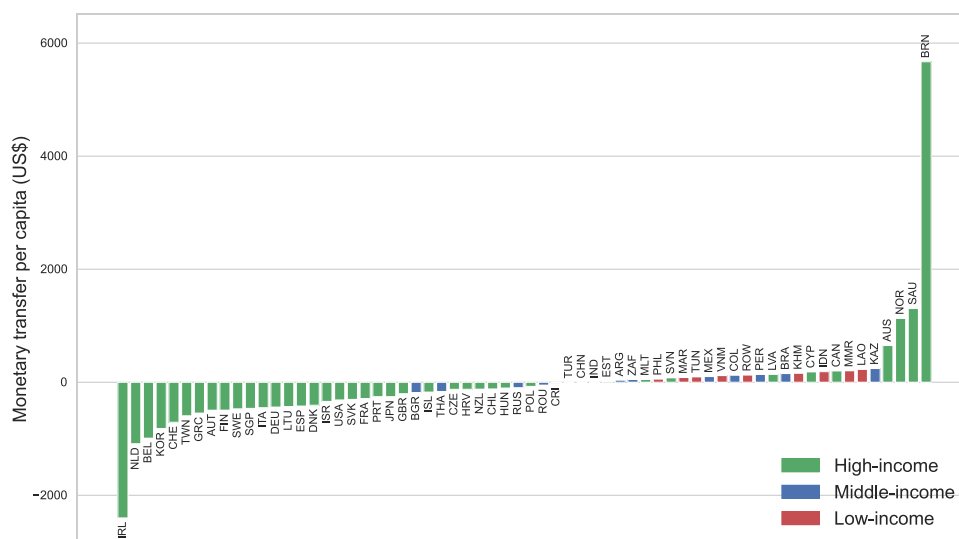
(a) Consumer tax



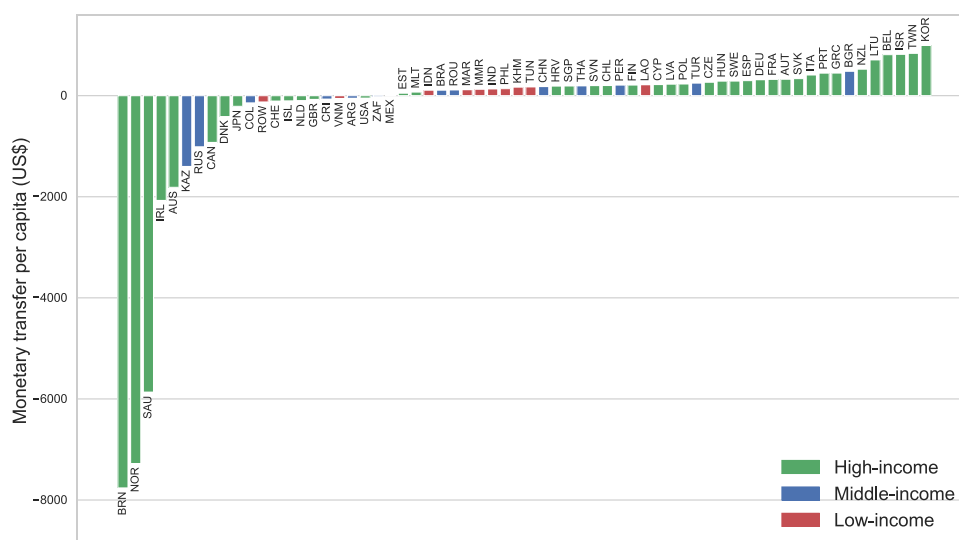
(b) Producer tax

Fig. 8. Change in total emissions intensities.

Note: This figure illustrates the average changes in total emission intensities across sectors following the implementation of a 100 USD/tCO₂e carbon tax. The *x*-axis is displayed on a log-scale. See text in Section 2.7 for a description of the formula used to calculate total emission intensities, which account for both direct emissions at the final production stage and cumulative emissions from all intermediate steps along the supply chain.



(a) Consumer tax (tax paid at destination)



(b) Producer tax (tax paid at origin)

Fig. 9. Equal-cost transfers by country under consumer and producer carbon taxes.

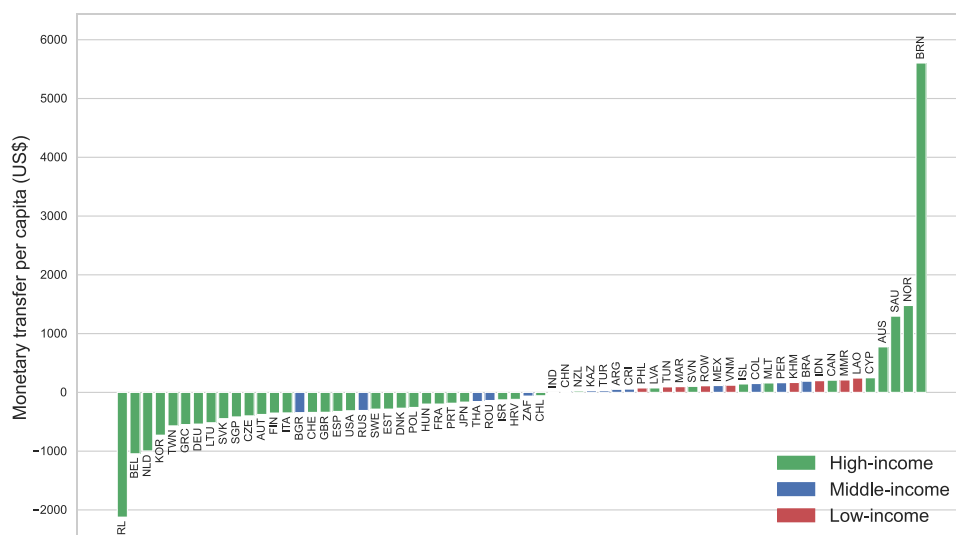
Note: This figure shows the *logarithm* of equal-cost transfers across countries required to compensate for real income losses under different carbon-tax regimes. Countries are grouped by income levels.

4.2.2. Historical polluter-pays transfers

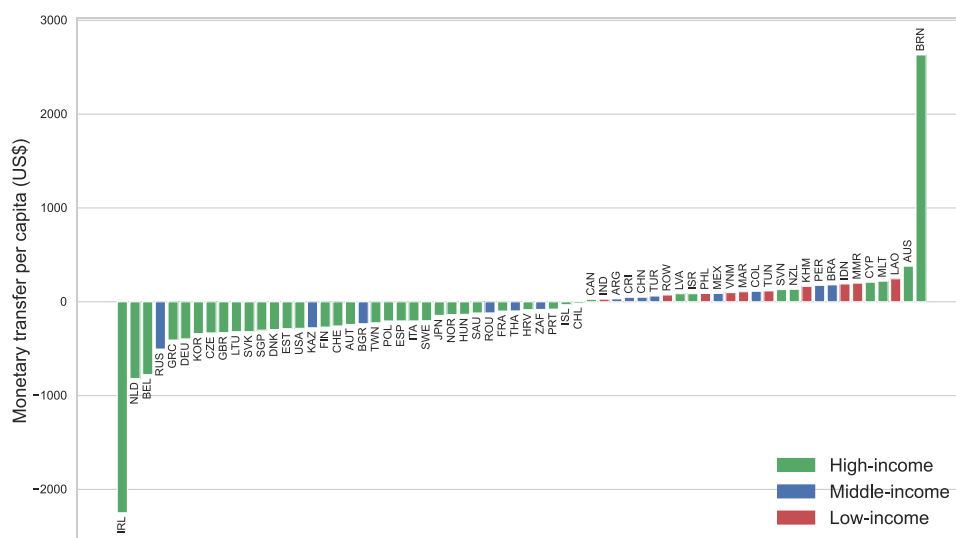
We now examine the international monetary transfers required under a polluter-pays scheme. This approach ensures that the real-income costs borne by each country are proportional to their historical contribution to global cumulative emissions.¹⁷ In this case, transfers align economic costs with historical emissions.¹⁸ For example, under this scheme the U.S. bears 25% of the aggregate economic costs, China bears 13%, and Brazil bears 1%, reflecting their respective contributions to total greenhouse gas emissions.

¹⁷ Fig. 18 in the Appendix displays historical cumulative emissions per capita for each country, based on data from Ritchie (2019). The data show that countries with the highest GDP per capita are also responsible for the largest share of historical emissions per capita, while low-income countries have historically contributed very little. Additionally, Fig. 19 in the Appendix shows that the U.S. accounts for approximately 25% of cumulative historical emissions, followed by China at around 14%.

¹⁸ Fig. 20 in the Appendix illustrates the resulting perfect correlation between changes in real income and each country's share of cumulative historical emissions.



(a) Consumer tax (tax paid at destination)



(b) Producer tax (tax paid at origin)

Fig. 10. Historical polluter-pays transfers by country under consumer and producer carbon taxes.

Note: This figure shows the *historical polluter-pays transfers* across countries under consumer taxation (panel a) and producer taxation (panel b). Transfers assign each country a burden proportional to its historical share of cumulative global emissions. Countries are grouped by income levels.

As in the equal-cost scenario, the polluter-pays scheme leads to similar aggregate outcomes compared to climate action without redistribution.

In terms of transfers, three key points emerge. First, the total volume of transfers under the polluter-pays rule is either very similar to the equal-cost scenario in the consumer-tax case (USD 237.3 billion) or substantially smaller in the producer-tax case (USD 201.3 billion). This difference reflects the distinct goals of the two schemes: the equal-cost rule equalizes economic losses across countries, whereas the polluter-pays rule allows losses to differ according to historical emissions. As a result, the polluter-pays allocation does not compensate many high-income countries for the real-income losses they incur under producer taxation—losses that, under the equal-cost rule, required large transfers from fossil-fuel exporters. Because these same high-income countries are also historically large emitters, they instead become net contributors. Consequently, per-capita transfers are generally smaller than under the equal-cost allocation, with the largest payment, slightly above USD 2000 per person per year for Ireland, remaining broadly similar across both tax regimes.

Second, the cross-country distribution of transfers continues to align strongly with income levels. With a few exceptions, most high-income economies appear as net payers and most low- and middle-income countries as net recipients, as shown in Fig. 10. This pattern reflects the tight link between contemporary GDP per capita and historical contributions to cumulative global emissions. Countries that industrialized early, emitting significant amounts of greenhouse gases since the 19th century, are now among the world's richest, while countries with low historical emissions remain in lower income brackets. This structural relationship drives the observed correlation between polluter-pays transfers and global income distribution. Notably, transfers from the Global North to the Global South are higher under the polluter-pays allocation: USD 173.3 billion in the consumer-tax case and USD 169.3 billion in the producer-tax case, corresponding to increases of 4.7% and 57% relative to the equal-cost scenario, respectively.

Third, under the polluter-pays rule the choice between consumer- and producer-tax regimes becomes far less consequential. In the equal-cost scenario, the location of taxation substantially influences the magnitude and direction of transfers, particularly because producer taxes generate large welfare gains for fossil-fuel exporters. Under the polluter-pays rule, however, transfers are determined chiefly by historical responsibility rather than by current tax incidence. Accordingly, the set of contributing and receiving countries – and the associated magnitudes – looks strikingly similar across the two regimes. This convergence highlights that once historical emissions are adopted as the normative criterion, the origin or destination of taxation plays only a secondary role in shaping global redistribution.

5. Heterogeneous carbon pricing

In this section, we examine the implications of heterogeneous carbon pricing schemes, where countries adopt varying levels of carbon taxation. These schemes are often considered more politically feasible, as they circumvent the need for universal coordination and mitigate the risk of free-riding. Rather than analyzing strategic incentives, we focus on evaluating the distributional impacts of these schemes by comparing them to a global policy benchmark. Our analysis centers on a prominent policy option currently shaping the policy debate: heterogeneous carbon prices based on income levels, as proposed by the International Monetary Fund (IMF) (Parry et al., 2021).

When investigating each scenario, we focus on the following question: does the proposed scheme introduce more or less fairness between the economic North and the economic South? Our analysis benchmarks these policies against a counterfactual scenario that achieves the same level of emissions reduction through a global carbon tax paired with “equal cost” transfers. By holding the emissions reduction constant, we can isolate the impact of price heterogeneity on fairness, by comparing the required transfers, and on efficiency, by evaluating the associated real income costs.

The International Carbon Price Floor (ICPF), proposed by the IMF in June 2021, aims to align near-term climate goals with credible policy actions (Parry et al., 2021). The proposal relies on two main components: (i) a core “club” of key large emitting countries to simplify negotiations, and (ii) a focus on establishing a minimum carbon price that each participating country must impose on their CO₂ emissions. The IMF identifies two possible configurations for the club: one comprising Canada, China, the European Union, Great Britain, India, and the United States, and another including all G20 countries. The IMF calibration suggests three floor prices: USD 75/tCO₂e for advanced economies, USD 50/tCO₂e for high-income emerging markets (EMEs), and USD 25/tCO₂e for low-income EMEs.

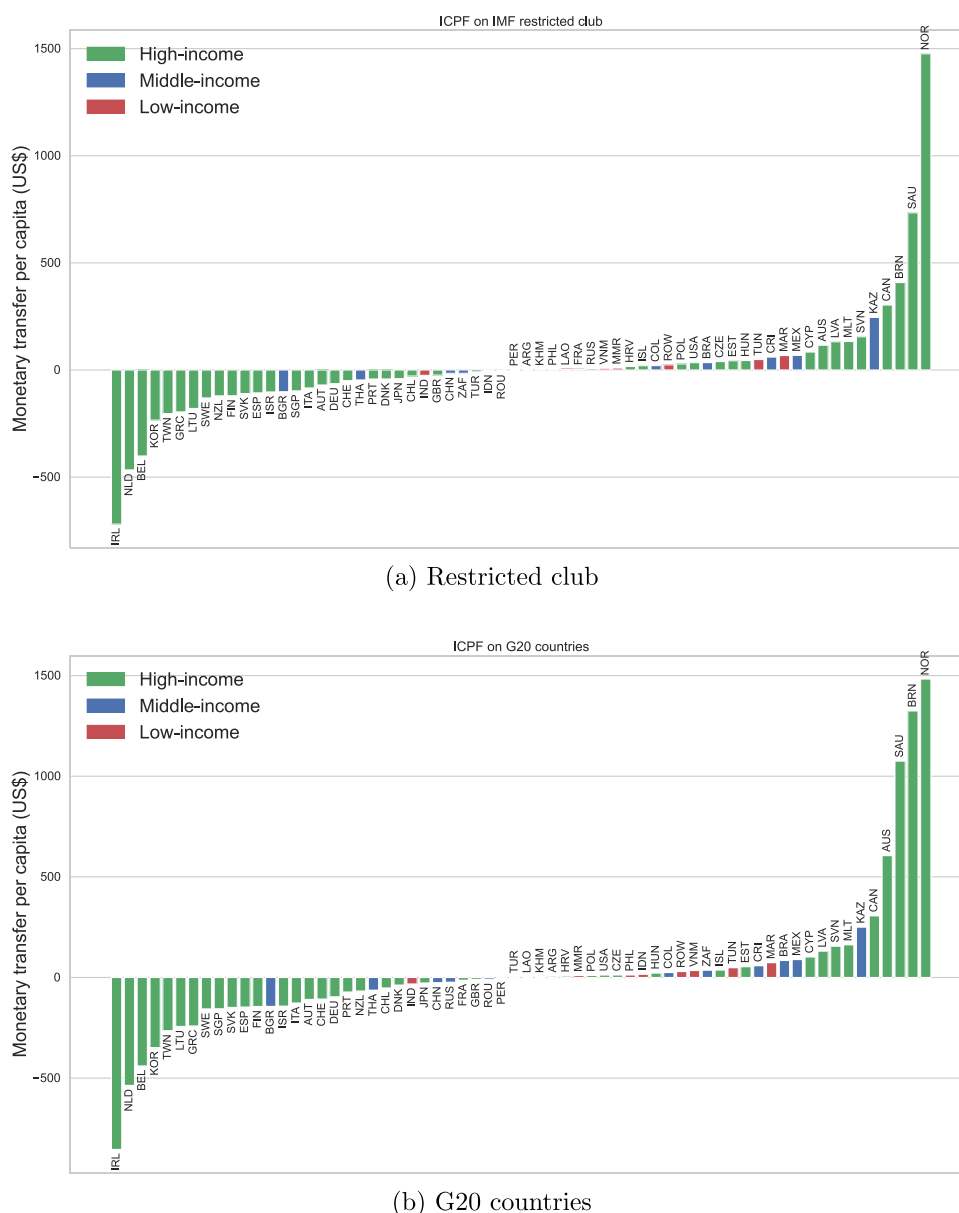
In our simulations, we analyze the two IMF-proposed scenarios (the restricted group and the G20), a scenario including “key players” from both the Global North and South: the EU, the U.S., China, India, Indonesia, Russia, Brazil, and South Africa, which jointly deliver more than two-thirds of the maximum emissions reduction achievable under a global uniform USD 100/tCO₂e carbon tax.¹⁹ Finally, we evaluate a version in which all countries participate in the ICPF, with carbon-price floors assigned according to income levels.²⁰ We compare the welfare costs and the ‘equal cost’ transfers in each ICPF scenario with those of a global carbon tax paired with transfers that achieve the same emissions reductions (see Figs. 11 and 12).

Our results show that the ICPF scenario including only Canada, China, the European Union, Great Britain, India, and the United States implies North–South transfers that are 8.7% higher than those required under a global uniform carbon tax. Expanding the ICPF to include all G20 countries or the “key players” club further increases this difference, with transfers 17% and 7% higher than the benchmark, respectively. When applied to all countries, however, the ICPF leads to slightly lower transfer needs, with a 3.7% reduction relative to the uniform tax. Figs. 11(a), 11(b), 12(a), and 12(b) illustrate the distribution of transfers in each scenario. As in the uniform-tax cases with equal-cost or polluter-pays transfers, high-income countries remain the main net payers. However, the correlation between transfers and income levels is weaker, even in the all-country ICPF scenario, because the policy targets income-based carbon price floors rather than directly addressing cross-country and cross-sector variation in emission intensities.

In terms of efficiency, all ICPF schemes generate higher global real-income costs than a uniform carbon tax. For the original IMF proposal (restricted club), the G20 scenario, and the “key players” club, the aggregate real-income loss is 1.68, 1.68 and 1.56 times larger respectively than under the benchmark uniform tax that achieves the same emissions reduction. The all-country ICPF,

¹⁹ This selection of countries is based on their significant contributions to emissions reductions under a uniform carbon tax, as illustrated in Fig. 26 in Appendix A. The figure shows the cumulative emissions reductions achieved as more countries adopt a carbon tax, progressing from the bottom to the top. When these key players alone implement a 100 USD/tCO₂e tax, a 30% reduction in global emissions is achieved—representing 76% of the maximum reduction possible with global adoption.

²⁰ For the all-country scenario, we use the World Bank's classification of countries by income level, available at: <https://datatopics.worldbank.org/world-development-indicators/the-world-by-income-and-region.html>. Fig. 21 in Appendix E shows how GDP per capita relates to heterogeneous carbon prices in this scenario.



(a) Restricted club

(b) G20 countries

Fig. 11. Equal-cost transfers by country under IMF ICPF proposals (consumer tax: restricted club and G20).

Note: This figure shows the *logarithm of equal-cost transfers* across countries under the IMF's International Carbon Price Floor. Panel (a) reports transfers for the restricted club scenario; panel (b) for the G20 group scenario. Transfers compensate countries for real income losses under the consumer-tax regime. Countries are grouped by income levels.

however, performs very differently: it leads to real-income costs that are about six times higher than the benchmark. This striking difference arises because in the more restricted schemes many low- and middle-income countries remain outside the tax regime. These countries either experience real income gains or face much milder losses, since they are not subject to carbon taxation. When all countries participate in the ICPF, by contrast, the policy applies globally, reducing these offsetting effects and generating a much stronger overall contraction in fossil-fuel use. As a result, the emissions reduction is also considerably larger—about 35%, compared with 26%, 20%, and 22% in the G20, restricted-club, and key-players scenarios, respectively.

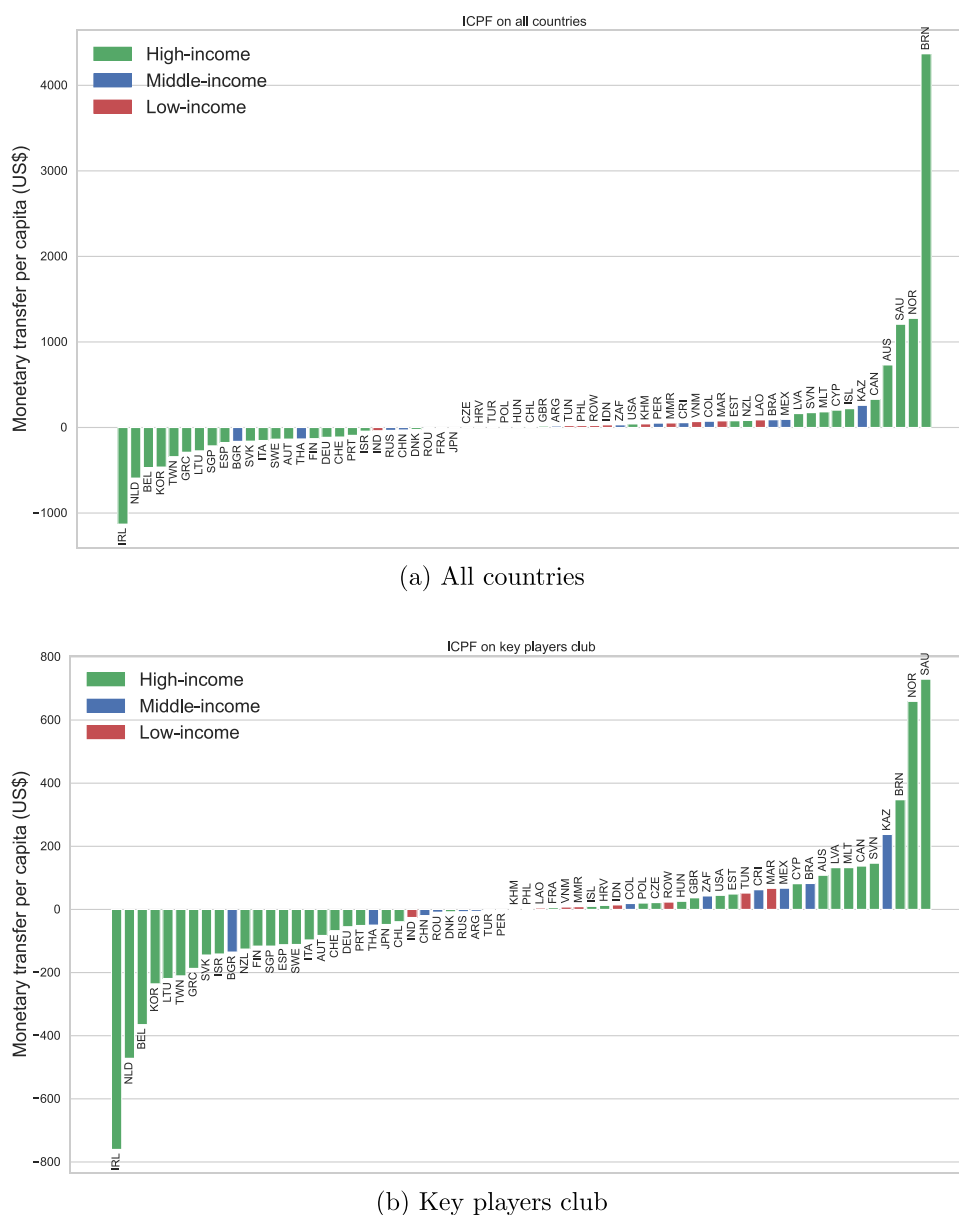
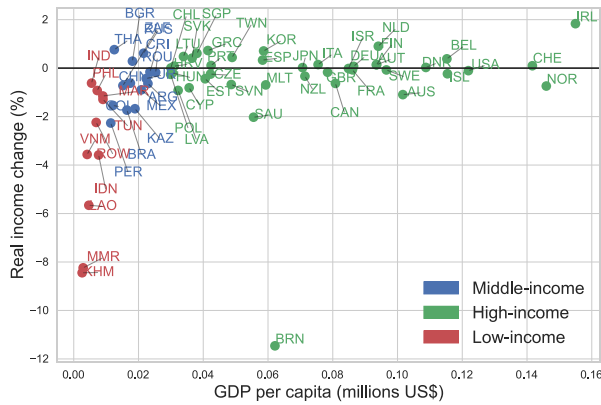


Fig. 12. Equal-cost transfers by country under IMF ICPF proposals (consumer tax: all countries and key players).

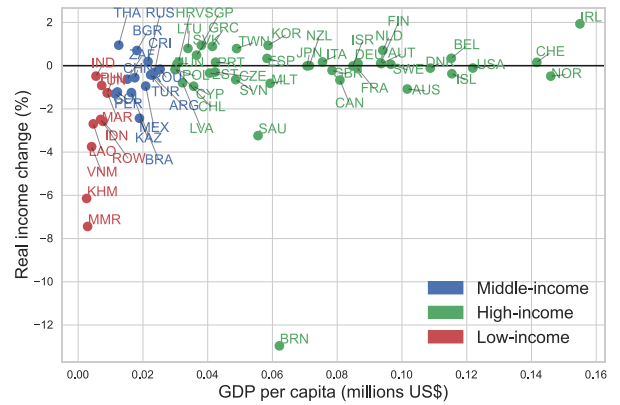
Note: This figure shows the *logarithm of equal-cost transfers* across countries under the IMF's International Carbon Price Floor. Panel (a) reports transfers for the all countries scenario; panel (b) for the key players club. Transfers compensate countries for real income losses under the consumer-tax regime. Countries are grouped by income levels.

Overall, our analysis suggests that the IMF proposal does not offer a clear advantage in fairness compared to a global uniform carbon tax paired with North-South transfers. Even in the best-case scenario, where all countries participate, the modest reduction in required transfers comes at the expense of higher real income costs.²¹

²¹ Figs. 22–25 in the Appendix present changes in real income by country for each scenario.

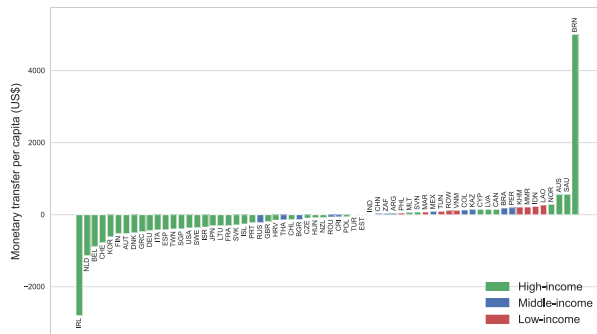


(a) Fontagné et al. estimation

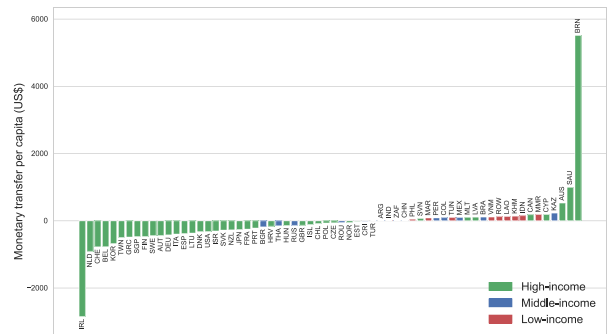


(b) Aggregated elasticities

Fig. 13. Real income changes under alternative elasticity estimates.

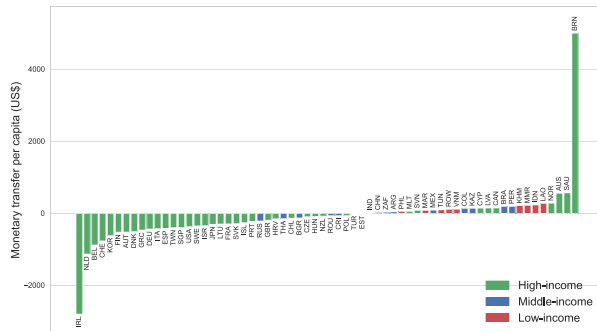


(a) Fontagné et al. estimation

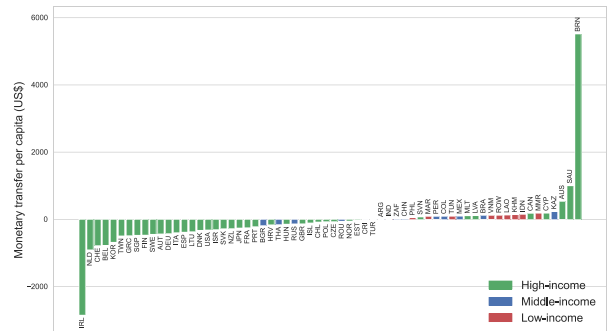


(b) Aggregated elasticities

Fig. 14. Equal-cost transfers under alternative elasticity estimates.



(a) Fontagné et al. estimation



(b) Aggregated elasticities

Fig. 15. Polluter-pays transfers under alternative elasticity estimates.

6. Robustness

In this section, we assess the robustness of our results by testing different values of the trade elasticity. Our main analysis relies on elasticities estimated using the method outlined in [Caliendo and Parro \(2015\)](#). To ensure the robustness of our findings, we also consider two alternative sets of elasticities. First, we adopt the elasticities provided in [Fontagné et al. \(2022\)](#), aggregated to match our sectoral definitions. Second, we apply the method in [Fontagné et al. \(2022\)](#) to estimate elasticities using our dataset,

supplemented with tariff data from Fontagné et al. (2022). Additional details regarding the estimation of elasticities are provided in Appendix D.

The results in Fig. 13 show the change in real income for each country under all three elasticity specifications. The qualitative pattern is unchanged: high-income countries generally experience real-income gains, while most low- and middle-income countries incur losses.

Figs. 14 and 15 report the corresponding equal-cost and polluter-pays transfers. Although the magnitudes vary somewhat across elasticity estimates, the core conclusions of the paper remain robust. In every case, transfer flows display a strong and systematic correlation with income levels: high-income countries are net contributors and low- and middle-income countries are net recipients, regardless of the elasticity dataset or the transfer rule applied.

7. Conclusion

This paper provides a comprehensive quantitative analysis of the distributional effects of carbon pricing across countries. Leveraging a modern, multi-country, multi-sector quantitative trade model, we investigate the implications of various carbon taxation schemes, identifying the inequities they create and proposing strategies to ensure fairness while maintaining economic efficiency. Our findings serve as critical benchmarks with immediate relevance for policymaking.

We find that while a uniform carbon tax effectively reduces global emissions, it disproportionately impacts low- and middle-income countries, exacerbating global economic inequalities. Redistribution mechanisms – whether aimed at equalizing the costs of climate action or addressing historical GHG emission patterns – could mitigate these disparities. Implementing such mechanisms would require tripling current climate finance levels, yet this translates into a per capita cost for high-income countries of 89–200 USD on average. Furthermore, we find that heterogeneous pricing schemes, such as the IMF's carbon taxation proposal, do not inherently produce fairer outcomes compared to global policies paired with redistribution.

This study highlights important directions for future research. Examining how technological innovation or diffusion, particularly in clean energy stimulated by carbon taxation, might influence the distributional impact of climate policies offers a promising avenue for further exploration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data treatment

Aggregations

To avoid sparseness of the input output table and zero gross outputs, we aggregate the following countries:

- Luxembourg and Belgium: subsequently labeled BEL in all data
- Hong-Kong and China: subsequently labeled CHN
- Malaysia and Singapore: subsequently labeled SGP

as well as the following sectors:

- 'Mining and quarrying, energy producing products' [D05T06] with 'Mining support service activities' [D09]: subsequently labeled as [D05T06] (Mining, energy) and later disaggregated into coal, crude oil, natural gas as detailed below
- 'Motor vehicles, trailers and semi-trailers' [D29] with 'Other transport equipment' [D30]: subsequently labeled [D29T30] (Transport equipments)

These aggregations leave us with a sample of 64 countries (incl. ROW aggregate) and 42 (after disaggregation 44) sectors from 1995 to 2018.

ICIO

The raw ICIO tables record negative values for some accounts of final consumption or value added. As the model cannot accommodate these negative values, we redistribute the negative parts in the table while respecting the following constraints:

- the sum of the columns and the sum of the rows must remain equal,
- the technical coefficients within the IO table (intermediate input spending over gross output ratio, corresponding to the parameters γ in the model) must remain constant equal to the raw ratios.

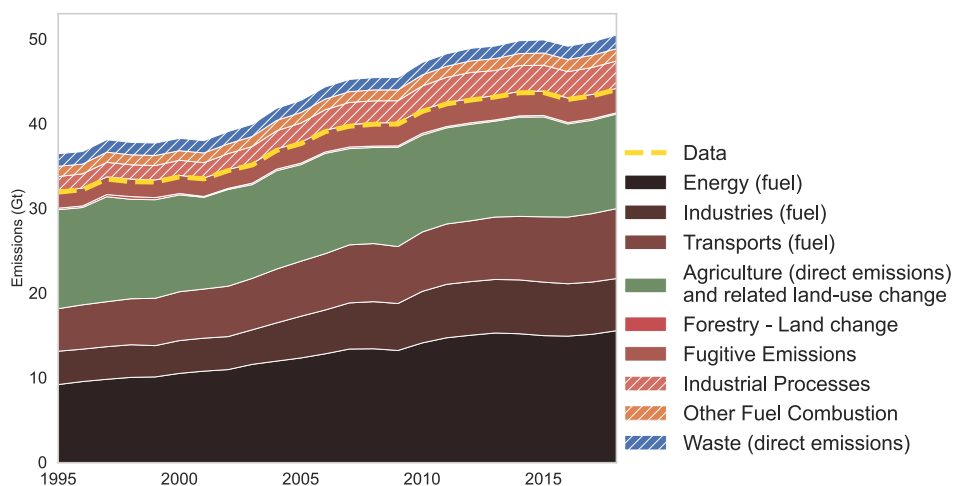


Fig. 16. Emissions by category.

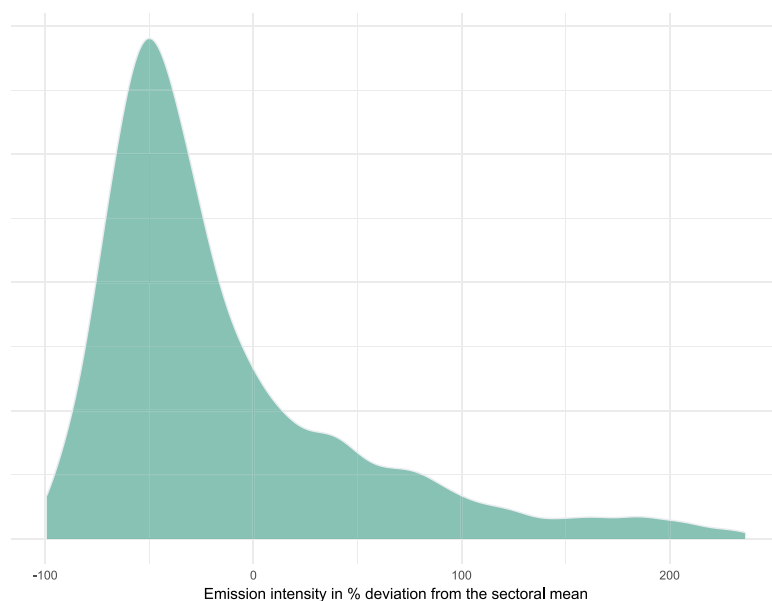


Fig. 17. Density of total emissions intensity.

Exiobase

As the ICIO tables only have one fossil fuel sector (mining, energy, code 05T06), we use the Exiobase input–output tables to disaggregate the ICIO fossil fuel sector into oil, gas, and coal (Stadler et al., 2025). We define the Exiobase sectors of “Anthracite”, “Coking Coal”, “Other Bituminous Coal”, “Sub-Bituminous Coal”, “Patent Fuel”, “Lignite/Brown Coal”, “BKB/Peat Briquettes”, and “Peat” as coal; “Crude petroleum and services related to crude oil extraction, excluding surveying” as oil; and “Natural gas and services related to natural gas extraction, excluding surveying”, “Natural Gas Liquids”, and “Other Hydrocarbons” as gas. We then obtain the output shares of the Exiobase oil, coal, and gas sectors using the Exiobase output file “x”. We proceed to disaggregate the ICIO data with the Exiobase shares for output, consumption, input–output, and value added.

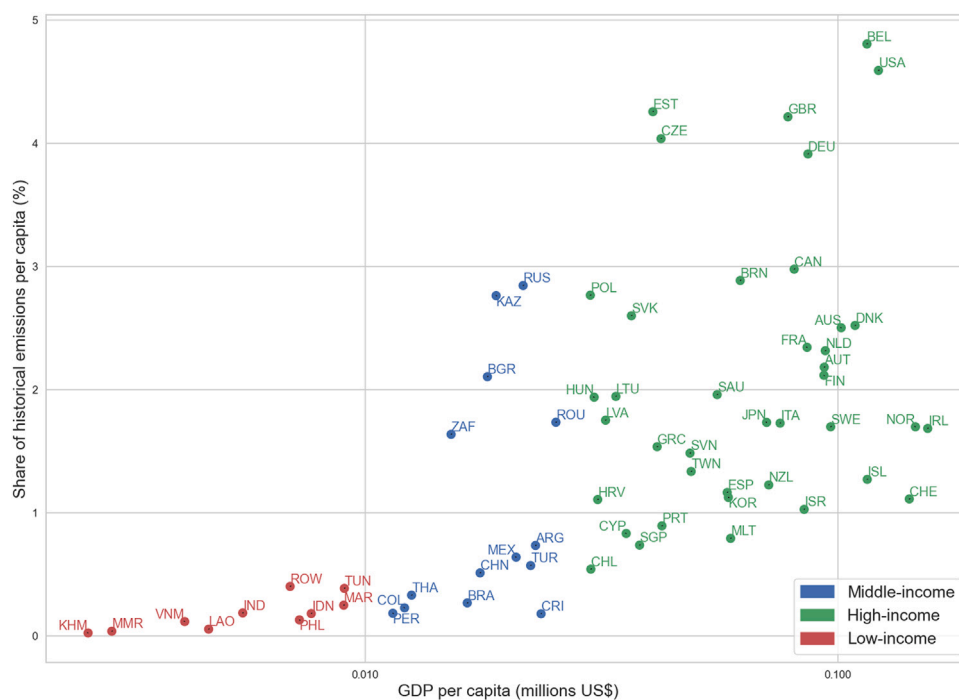


Fig. 18. Historical emissions.

Source: Ritchie (2019).

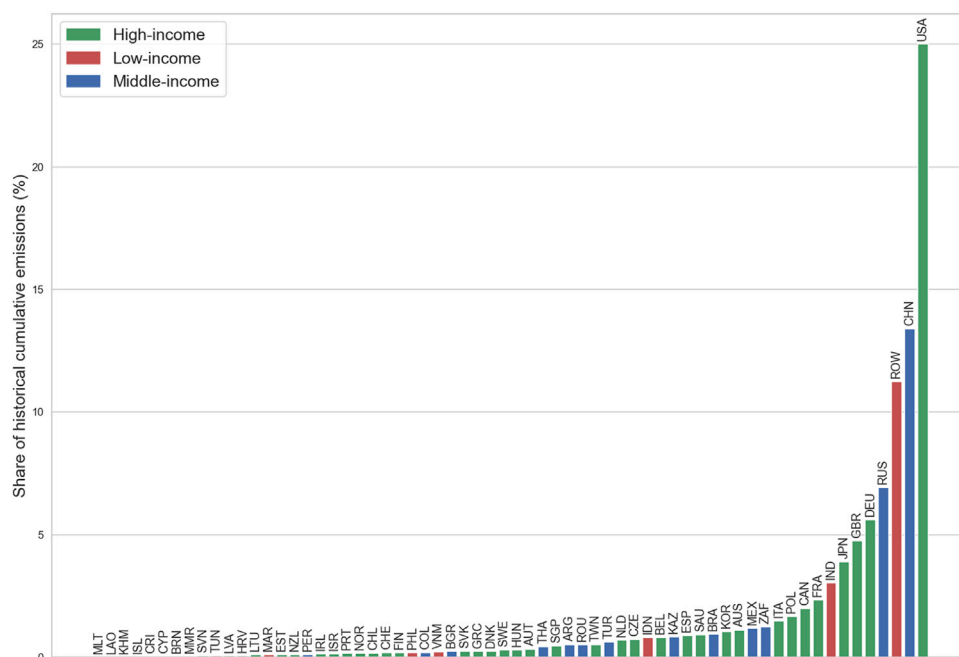


Fig. 19. Cumulative historical emissions.

Source: Ritchie (2019).

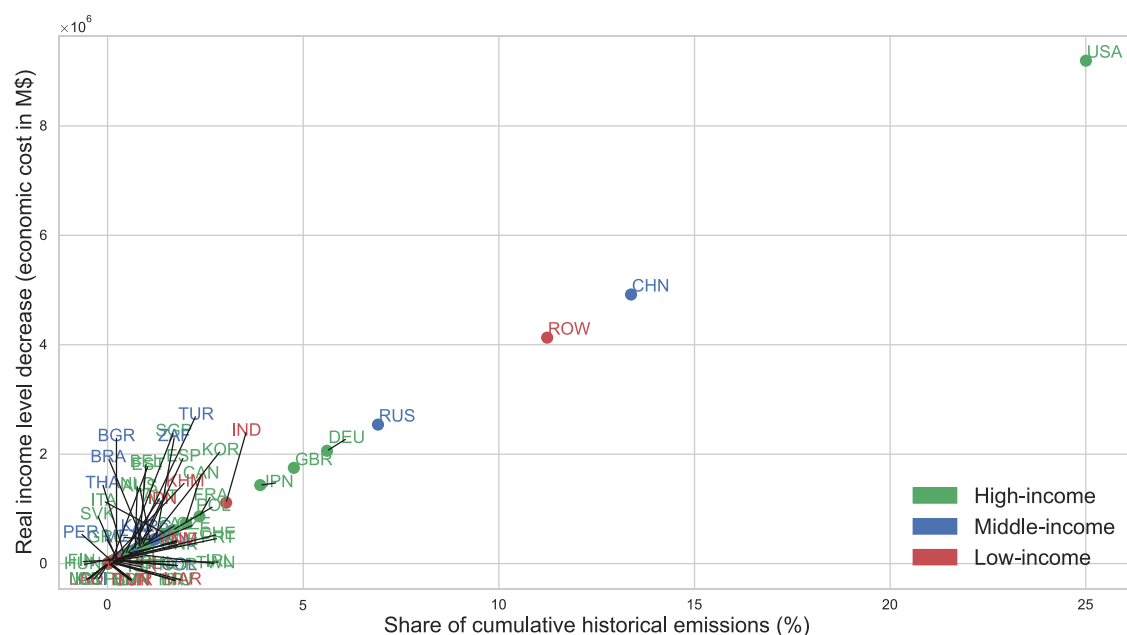


Fig. 20. Historical polluter pays transfers: Changes in real income by country.

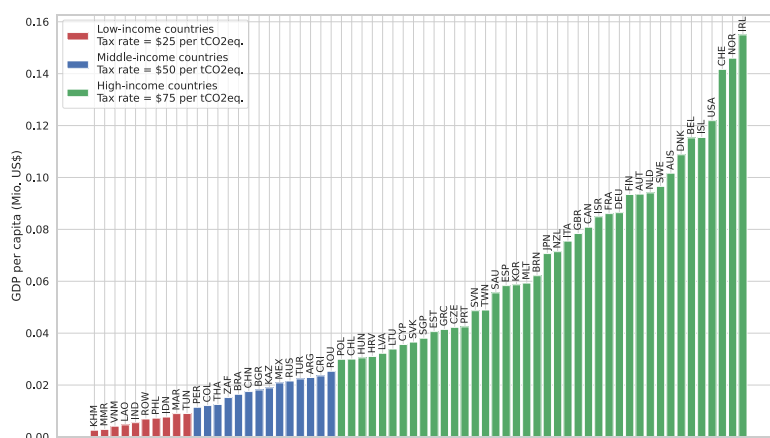


Fig. 21. IMF proposal.

FAO

We keep only FAO Scope 1 emissions by subcategories belonging to the category ‘Agricultural Land’ with the exception of ‘On-farm Energy Use’, since these emissions are contained in the emissions from the fuel sectors.²² The remaining observations are then aggregated into the 64 countries with the ROW aggregate and represent the emissions intensities of the agricultural sector.

For the agricultural price index, we use producer prices in USD/tonne for 2018 where available. For Myanmar, the producer prices are only available in local currency, so we use FAO’s exchange rate to convert the prices to USD. We use quantities from the FAO crops and livestock products statistics to create a weighted price index of agricultural production. As the ICIO country definition is different from the FAO’s, we use GDP data from the FAO macroeconomic indicators to aggregate the countries to the ICIO definition.

²² The category ‘Agricultural Land’ includes the following subcategories: ‘Fires in humid tropical forests’, ‘Fires in organic soils’, ‘Net Forest conversion’, ‘Drained organic soils’, ‘Synthetic Fertilizers’, ‘Crop Residues’, ‘Manure left on Pasture’, ‘Manure applied to Soils’, ‘Manure Management’, ‘Enteric Fermentation’, ‘Savanna fires’, ‘Burning - Crop residues’, ‘Rice Cultivation’, ‘On-farm Energy Use’.

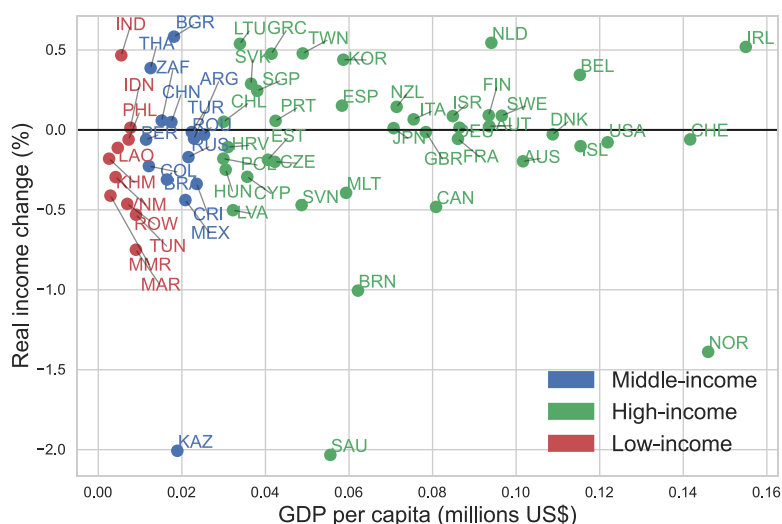


Fig. 22. Real income changes - IMF proposal - restricted club.

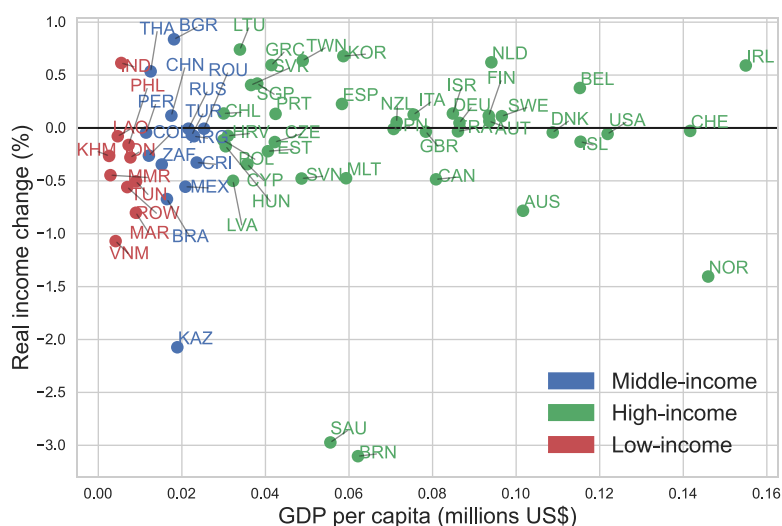


Fig. 23. Real income changes - IMF proposal - G20.

IEA

We obtain cross-country data on fossil-fuel prices from the *IEA World Energy Balances Highlights 2024*, sheet *TimeSeries_1971–2023*. We extract all observations for which *Flow* equals *Production (PJ)* and the product corresponds to one of the three major fossil fuels: coal, crude oil (including NGL and feedstocks), and natural gas. We retain only observations for 2018, harmonize country names using ISO-3 codes. For each fuel, we construct a Rest-of-World entry as the difference between the global total and the sum across reporting countries.

Energy quantities are converted from petajoules into tonnes of oil equivalent using the factor $1 \text{ PJ} = 23,884.59 \text{ toe}$. Each fuel category is mapped to its corresponding production sector, and the energy dataset is merged with monetary production values. We then compute the fossil-fuel prices as the ratio of monetary value to physical production for each country. Missing values are replaced by sectoral means, and we extend the dataset to ensure full coverage for all 64 countries in the model. The final output is a country-by-fuel price matrix consistent with observed dispersion in extraction costs and segmentation across global energy markets.

EDGAR

We first combine different time series extracts of the EDGAR database, namely the 'CH4', 'CO2_excl_short-cycle_org.C' and 'N2O' data sheet by converting the emissions into CO₂ equivalents according to the respective AR4 100-year GWP value.²³ We then

²³ The AR4 100-year GWP values are 25 for CH₄ and 298 for N₂O.

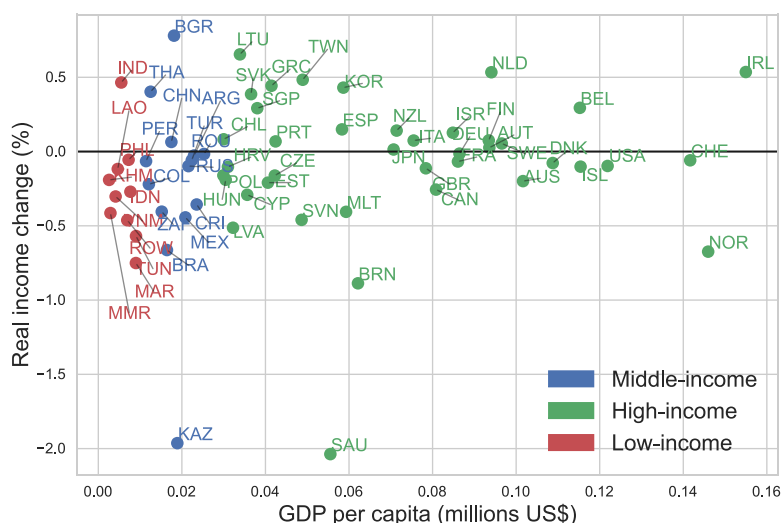


Fig. 24. Real income changes - IMF proposal - key players.

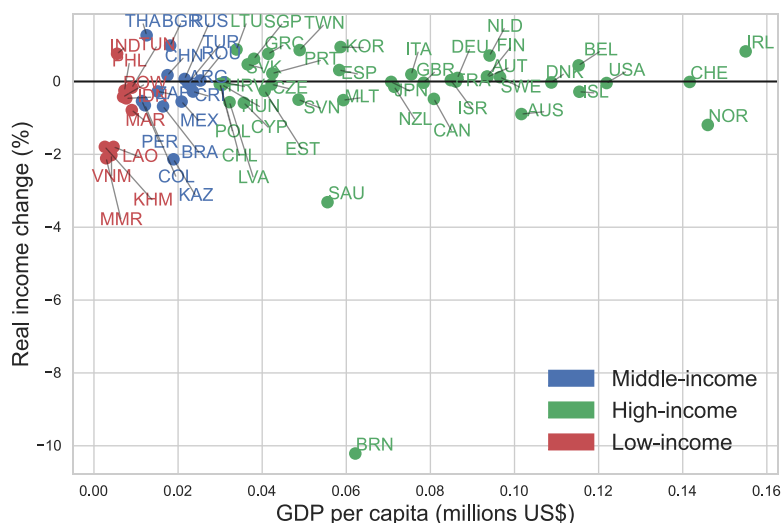


Fig. 25. Real income changes - IMF proposal - all countries.

aggregate the data into our 64 sample countries and create the ROW aggregate with the remaining countries. To assign the IPCC emission categories to our various sample sectors, we rely on the exact definition of the IPCC emission category compared to the ISIC rev.4 codes comprised in our sample sector definition. We allocate the IPCC category “fugitive emissions” (chapter 1.B) to the mining, energy sector which is then disaggregated into coal, natural gas, and crude oil according to output shares.

Appendix B. Data description

Fig. 16 shows the evolution of emissions according to their source, again in Gigatons of CO₂e. The brown areas on the bottom of the graph (energy, industries, transports) relate to fuel combustion. This graph shows why including methane and nitrous oxide emissions is important: Emissions related to agriculture are a large share of the overall emissions. If we only considered emissions from fuel combustion, the impact of the agricultural sector on greenhouse gas emissions would be underestimated.

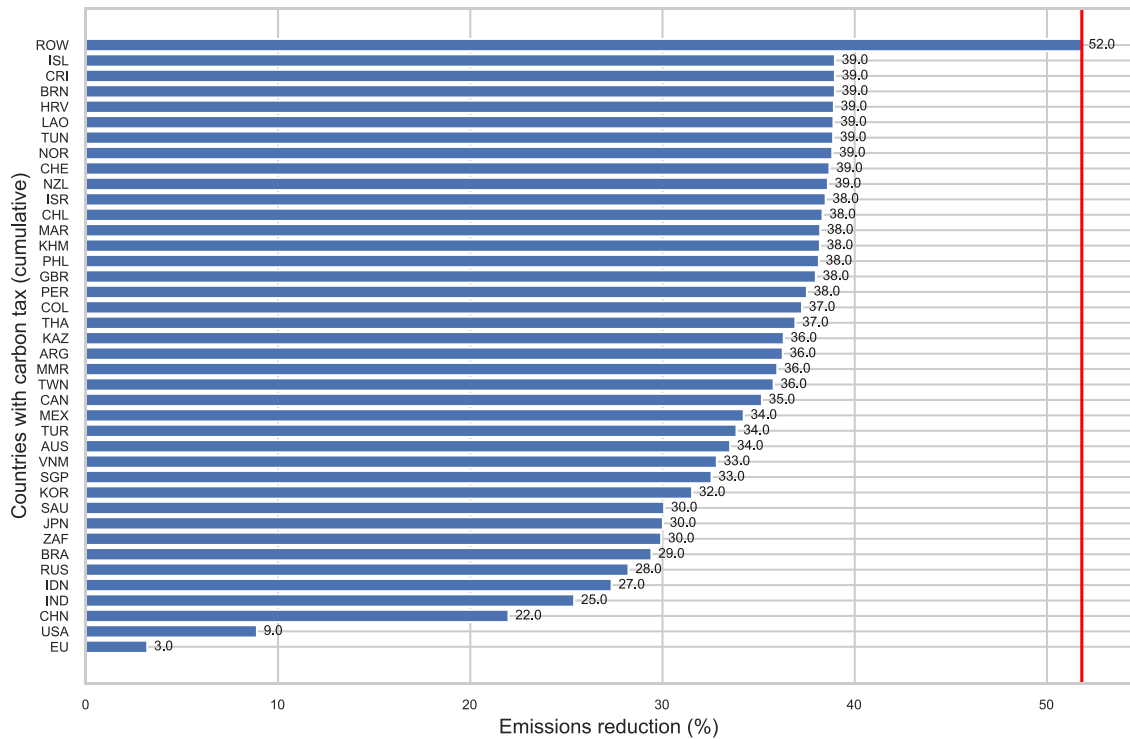


Fig. 26. Incremental carbon taxes.

Note: This figure depicts emissions reduction when imposing a uniform carbon tax of USD100/ton of CO₂ on an incrementally larger set of countries. The x-axis depicts the emissions reduction, while the y-axis indicates which country is newly added to the scenario (from the bottom to the top).

Appendix C. Solving algorithm

In this section, we detail how we reduce the model to a $N \times S$ system that we use to back out counterfactual results. The aim is to reduce the number of variables, while keeping the system as linear as possible.

Eqs. (11) and (12) imply that :

$$\tilde{q}_{is'j} = \tilde{I}_j \tilde{q}_{is'j}^{\circ} \quad (27)$$

where $\tilde{q}_{is'j}^{\circ} = [\tilde{p}_{is'}(1 + t_{is'j}^e)]^{-\sigma_{s'j}} \tilde{P}_{s'j}^{\sigma_{s'j}-1}$ only depends on the change of prices and the baseline. Useful to note that its inverse is linear in any change of set of prices : $\tilde{q}_{is'j}^{\circ}(\alpha \tilde{p}) = \tilde{q}_{is'j}^{\circ}(\tilde{p})/\alpha$. This is the most general expression of the change of quantities traded so that the condition of consumer spending is respected by construction because : $\sum_{i,s'} \tilde{p}_{is'}(1 + t_{is'j}^e) \tilde{q}_{is'j}^{\circ}(p_{is'j}^B, q_{is'j}^B) = I_j^B$. The form of this expression represents that if the income of the consumer increases (or decreases), they will proportionally increase their consumption from every country/sector. $\tilde{q}_{is'j}^{\circ}$ contains all the information of the reorganization of their consumption if their income did not change in the counterfactual world.

Similarly, for intermediates, Eq. (13) together with $\tilde{E}_{s'js} = \tilde{E}_{js}$ imply that :

$$\tilde{m}_{is'js} = \tilde{E}_{js} \tilde{m}_{is'js}^{\circ} \quad (28)$$

where $\tilde{m}_{is'js}^{\circ} = [\tilde{p}_{is'}(1 + t_{is'j}^e)]^{-\eta_{s'j}} \tilde{P}_{s'js}^{\pi(\eta_{s'j}-1)}$ has the same properties as $\tilde{q}_{is'j}^{\circ}$. The construction makes sure that the producer spending is respected.

Having the consumer and producer spending respected by construction, we need to compute the consumer and producer revenue. We use Eqs. (20) with $\tilde{E}_{js} = \tilde{Y}_{js}$ to compute the wages change under a change of spending of the producer:

$$\tilde{w}_j = \frac{\sum_s \tilde{E}_{js} L_{js}^B}{L_j} \quad (29)$$

We have then made sure that the solution respects the labor market clearing condition and the constitutive equation of production $L_{js} = \gamma_{j,Ls} \frac{I_{js}}{w_j}$, and we can write the consumer revenue and producer spending from the consumer and producer clearing equations

under each tax regime. In the case of the consumer tax, we have:

$$\tilde{I}_j I_j^B = \sum_s \tilde{E}_{js} L_{js}^B w_j^B + \sum_s \tilde{E}_{js} Y_{js}^B + \sum_{i,s'} \tilde{I}_j \tilde{p}_{is'} t_{is'}^e \tilde{q}_{is'}^o (p_{is'}^B q_{is'}^B) + \sum_{i,s',s} \tilde{E}_{js} \tilde{p}_{is'} t_{is'}^e \tilde{m}_{is'js}^o (p_{is'}^B m_{is'js}^B) + D_j^B \quad (30)$$

In the case of the producer tax, we have:

$$\tilde{I}_j I_j^B = \sum_s \tilde{E}_{js} L_{js}^B w_j^B + \sum_{i,s'} \tilde{I}_j \tilde{p}_{js'} t_{js'i}^e \tilde{q}_{js'i}^o (p_{js'i}^B q_{js'i}^B) + \sum_{i,s',s} \tilde{E}_{is} \tilde{p}_{js'} t_{js'i}^e \tilde{m}_{js'is}^o (p_{js'i}^B m_{js'is}^B) + D_j^B \quad (31)$$

And the producer equation is given by:

$$\tilde{E}_{is'} E_{is'}^B = \sum_j \tilde{I}_j \tilde{p}_{is'} \tilde{q}_{is'j}^o (p_{is'j}^B q_{is'j}^B) + \sum_{j,s} \tilde{E}_{js} \tilde{p}_{is'} \tilde{m}_{is'js}^o (p_{is'j}^B m_{is'js}^B) \quad (32)$$

We use the equations above to reduce the system to a parsimonious set of equations. In the consumer tax case, we use (30) in (32):

$$\tilde{E}_{is'} E_{is'}^B = \tilde{p}_{is'} \left(\sum_{j,s} \tilde{E}_{js} \left[\tilde{q}_{is'j}^o (p_{is'j}^B q_{is'j}^B) \frac{L_{js}^B w_j^B + Y_{js}^B + \sum_{i,s'} \tilde{m}_{is'js}^o \tilde{p}_{is'} t_{is'}^e (p_{is'j}^B q_{is'j}^B)}{I_j^B - \sum_{i,s'} \tilde{q}_{is'j}^o \tilde{p}_{is'} t_{is'}^e (p_{is'j}^B q_{is'j}^B)} + \tilde{m}_{is'js}^o (p_{is'j}^B m_{is'js}^B) \right] + \sum_j \frac{D_j \tilde{q}_{is'j}^o (p_{is'j}^B q_{is'j}^B)}{I_j^B - \sum_{i,s'} \tilde{p}_{is'} t_{is'}^e \tilde{q}_{is'j}^o (p_{is'j}^B q_{is'j}^B)} \right) \quad (33)$$

$$\tilde{p}_{js} = \left(\sum_{s'} \tilde{E}_{js'} \frac{L_{js'}^B}{L_j} \right)^{\gamma_{j,LS}} \prod_{s'} \tilde{p}_{js'}^{\pi_{\gamma_{js'}}} \quad , s \notin S^M \quad (34)$$

$$\tilde{E}_{j,s} = (\tilde{p}_{js})^{\frac{1}{\gamma_{j,Ms}}} \left(\sum_{s'} \frac{\tilde{E}_{js'} L_{js'}^B}{L_j} \right)^{-\frac{\gamma_{j,LS}}{\gamma_{j,Ms}}} \prod_k (\tilde{p}_{kjs}^{\pi_{\gamma_{kjs}}})^{-\frac{\gamma_{kjs}}{\gamma_{j,Ms}}} \quad , s \in S^M \quad (35)$$

with the last two equations expressing the cost of production from the solution of the cost minimization of the production costs of the producer (15). We have thus reduced the equations to a system of two non-linear Eqs. (33) and (34) of the two fundamental hat quantities $(\tilde{E}, \tilde{p})$. Since we have explicit expressions of the variables on the right hand side, we can solve numerically this system with a nested fixed point routine. The case of the producer tax follows analogously.

The solution space of this system of equations is of dimension 1, any linear transformation $\alpha(\tilde{E}_{sol}, \tilde{p}_{sol})$ of a solution of the system is also solution. We need to add one numeraire constraint to make the solution unique.

$$\tilde{X}_{numeraire} = 1 \quad (36)$$

The numeraire can be any quantity constructed with a linear combination of $(\tilde{E}_{sol}, \tilde{p}_{sol})$. We use the wage in the USA as a numeraire, so we need to enforce that $\tilde{E}_{sol}$ respects :

$$\sum_{s'} \tilde{E}_{USA,s'} \frac{L_{USA,s'}^B}{L_{USA}} = 1 \quad (37)$$

So we can solve for any solution $(\tilde{E}_{sol}, \tilde{p}_{sol})$ of (33) and (34) and construct :

$$\left(\frac{\tilde{E}_{sol}}{\sum_{s'} \left(\tilde{E}_{sol,USA,s'} \frac{L_{USA,s'}^B}{L_{USA}} \right)}, \frac{\tilde{p}_{sol}}{\sum_{s'} \left(\tilde{E}_{sol,USA,s'} \frac{L_{USA,s'}^B}{L_{USA}} \right)} \right) \quad (38)$$

that is the unique solution that respects (33), (34) and (37). Our results are robust to other choices of numeraire, such as global GDP or total global output.

Appendix D. Elasticity estimation

Table D.17 shows the elasticities for the different estimation methods. Most methods to estimate elasticities rely on the gravity equation, which maps trade flows to trade costs like distance and tariffs. All results in the main part of the paper are based on the elasticities in column 3 which we estimate along the lines of [Caliendo and Parro \(2015\)](#). This method uses a fixed-effects strategy to isolate the effect of trade costs on trade flows. We use all years of the data to create the fixed-effects following [Caliendo and Parro \(2015\)](#). The fifth column relies on the estimation method in [Fontagné et al. \(2022\)](#), where we use lagged tariffs as instruments for the tariffs in a regression of the gravity equation. For that, we use the tariff data in [Fontagné et al. \(2022\)](#), aggregate it to our level of sectoral aggregation, and combine it with our trade flow data. To estimate the elasticities, we regress log tariffs on log trade flows, together with multilateral resistance terms, importer-year, and exporter-year fixed effects. Three-year lags of tariffs serve as instruments for the tariffs. Column 7 shows the simplest method we use: We take existing elasticities and map them to our sector categorization. For that, we use the elasticities in [Fontagné et al. \(2022\)](#) and aggregate them so that on average, they yield the number 4 which is in line with the preferred elasticity in [Head and Mayer \(2014\)](#). The aggregation is necessary because we have a much less granular sectoral composition than [Fontagné et al. \(2022\)](#). We replace elasticities that have the wrong sign, are insignificant, or cannot be estimated due to a lack of tariff data by the mean.

Table D.17
Elasticities.

Sector	Sector names	CP elasticity	CP SE	FG elasticity	FG SE	Agg elasticity
01T02	Agriculture	4.95	0.14	8.27	0.69	2.80
03	Fishing	2.58	0.10	2.47	0.46	3.18
05T06.1	Coal	0.65	0.20	2.75	0.91	6.52
05T06.2	Oil	3.26	0.32	6.01	1.30	6.52
05T06.3	Gas	1.09	0.17	2.67	0.77	6.52
07T08	Mining, non-energy	2.82	0.22	4.03	1.00	8.58
10T12	Food products	4.05	0.14	6.31	0.44	2.84
13T15	Textiles	5.17	0.14	5.75	0.56	2.78
16	Wood	5.02	0.13	5.69	0.58	4.47
17T18	Paper	4.54	0.14	6.86	0.62	4.64
19	Coke, petroleum	3.51	NA	4.98	NA	4.67
20	Chemicals	3.77	0.19	6.72	0.85	4.73
21	Pharmaceuticals	4.10	0.23	5.33	1.08	7.58
22	Plastics	4.17	0.13	6.19	0.55	3.48
23	Non-metallic minerals	3.68	0.16	5.36	0.73	3.16
24	Basic metals	5.86	0.24	7.19	1.14	5.70
25	Fabricated metals	4.87	0.14	5.89	0.66	2.28
26	Electronic	2.19	0.16	5.63	0.72	2.69
27	Electrical equipment	3.24	0.13	6.00	0.58	2.34
28	Machinery	2.91	0.14	6.02	0.63	4.35
29T30	Transport equipments	3.51	NA	4.98	NA	4.86
31T33	Manufacturing nec	4.90	0.14	6.56	0.61	3.83
35	Energy	3.80	0.13	1.69	0.47	9.04
36T39	Water supply	1.91	0.21	5.05	1.12	6.16
41T43	Construction	3.51	NA	4.98	NA	2.59
45T47	Wholesale, retail	3.51	NA	4.98	NA	2.59
49	Land transport	3.51	NA	4.98	NA	2.59
50	Water transport	3.51	NA	4.98	NA	2.59
51	Air transport	3.51	NA	4.98	NA	2.59
52	Warehousing	3.51	NA	4.98	NA	2.59
53	Post	3.51	NA	4.98	NA	2.59
55T56	Tourism	3.51	NA	4.98	NA	2.59
58T60	Media	4.03	0.17	5.15	0.81	4.58
61	Telecom	3.51	NA	4.98	NA	2.59
62T63	IT	3.51	NA	4.98	NA	2.59
64T66	Finance, insurance	3.51	NA	4.98	NA	2.59
68	Real estate	3.51	NA	4.98	NA	2.59
69T75	R&D	3.76	0.15	2.38	0.71	4.81
77T82	Administration	2.46	0.10	2.62	0.55	4.70
84	Public sector	3.51	NA	4.98	NA	2.59
85	Education	2.80	0.11	2.78	0.52	4.65
86T88	Health	3.51	NA	4.98	NA	2.59
90T93	Entertainment	2.29	0.11	3.22	0.56	4.70
94T98	Other service	3.51	NA	4.98	NA	2.59

* This table shows different versions of the elasticity of substitution across origins. The sectors correspond to the OECD ICIO sectoral classification. We replace insignificant, negative, and nonestimable elasticities with the mean. These elasticities have no standard error.

Appendix E. Additional figures

See Figs. 17 and 26.

Data availability

The Distributional Effects of Carbon Pricing - Replication Package (Original data) (Mendeley Data)

References

- Adler, C., Wester, P., Bhatt, I., Huggel, C., Insarov, G., Morecroft, M., Muccione, V., Prakash, A., 2022. Cross-chapter paper 5: Mountains. In: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II To the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, UK and New York, USA, pp. 2273–2318.
- Aichele, R., Felbermayr, G., 2012. Kyoto and the carbon footprint of nations. J. Environ. Econ. Manag. 63 (3), 336–354, URL <https://www.sciencedirect.com/science/article/pii/S0095069611001422>.

- Armington, P.S., 1969. The geographic pattern of trade and the effects of price changes. *IMF Econ Rev* 16 16, 179–201, Symposium on Growth and International Trade: Empirical Studies.
- Babiker, M.H., 2005. Climate change policy, market structure, and carbon leakage. *J. Int. Econ.* 65 (2), 421–445.
- Blanchard, O., Gollier, C., Tirole, J., 2023. The portfolio of economic policies needed to fight climate change. "Annual Rev. Economics" 15 (Volume 15, 2023), 689–722, URL <https://www.annualreviews.org/content/journals/10.1146/annurev-economics-051520-015113>.
- Böhringer, C., Carbone, J., Rutherford, T., 2016. The strategic value of carbon tariffs. *Am. Econ. J.: Econ. Policy* 8, 28–51.
- Boyce, J.K., 2018. Carbon pricing: effectiveness and equity. *Ecol. Econom.* 150, 52–61.
- Caliendo, L., Dolabella, M., Moreira, M., Murillo, M., Parro, F., 2024. Voluntary Emission Restraints in Developing Economies: The Role of Trade Policy. National Bureau of Economic Research, Working Paper 32459. URL <http://www.nber.org/papers/w32459>.
- Caliendo, L., Parro, F., 2015. Estimates of the trade and welfare effects of NAFTA. *Rev. Econ. Stud.* 82 (1 (290)), 1–44.
- Calvacanti, T., Hasna, Z., Santos, C., 2024. Climate change mitigation policies: aggregate and distributional effects. *Econ. J.* 135 (668), 1341–1387.
- Copeland, B.R., 1996. Pollution content tariffs, environmental rent shifting, and the control of cross-border pollution. *J. Int. Econ.* 40 (3–4), 459–476.
- Dekle, R., Eaton, J., Kortum, S., 2007. Unbalanced trade. *Am. Econ. Rev.* 97 (2), 351–355.
- Elliott, J., Foster, I., Kortum, S., Khun Jush, G., Munson, T., Weisbach, D., 2013. Unilateral carbon taxes, border tax adjustments, and carbon leakage. *Theoret. Inq. Law* 14, 207–244.
- Elliott, J., Foster, I., Kortum, S., Munson, T., Cervantes, F.P., Weisbach, D., 2010. Trade and carbon taxes. *Am. Econ. Rev.* 100 (2), 465–469, URL <http://www.jstor.org/stable/27805040>.
- European Commission, J., 2023. Emission database for global atmospheric research (EDGAR). URL <https://edgar.jrc.ec.europa.eu/>. Joint Research Centre (JRC)/Netherlands Environmental Assessment Agency (PBL), release version 4.tox2 global gridmaps.
- FAO, 2023a. FAOSTAT agricultural producer prices, crops and livestock products, exchange rates, macroeconomics indicators. <https://data.apps.fao.org/catalog/dataset/faostat-pp>. (Accessed 18 November 2025).
- FAO, 2023b. FAOSTAT emissions totals. URL <https://www.fao.org/faostat/en/#data/GT>. License: CC BY-NC-SA 3.0 IGO. (Accessed 15 April 2024).
- Farrokhi, F., Lashkaripour, A., 2025. Can trade policy mitigate climate change? *Econometrica* 93 (5), 1561–1599, URL <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA20153>.
- Farrokhi, F., Lashkaripour, A., Taheri, H., 2025. A framework for integrating climate goals into trade agreements. Working Paper.
- Felder, S., Rutherford, T.F., 1993. Unilateral CO2 reductions and carbon leakage: The consequences of international trade in oil and basic materials. *J. Environ. Econ. Manag.* 25 (2), 162–176, URL <https://www.sciencedirect.com/science/article/pii/S0095069683710405>.
- Finkelstein-Shapiro, A., Nuguer, V., 2023. Climate policies, labor markets, and macroeconomic outcomes in emerging economies. *Labor Mark. Macroecon. Outcomes Emerg. Econ.* (January 19, 2023).
- Fontagné, L., Guimbard, H., Orefice, G., 2022. Tariff-based product-level trade elasticities. *J. Int. Econ.* 137, 103593.
- Garcia-Lembergman, E., Ramondo, N., Rodríguez-Clare, A., Shapiro, J.S., 2025. Carbon emissions in the global economy. URL <https://www.nber.org/>. Unpublished manuscript, July 2025.
- Golosov, M., Hassler, J., Krusell, P., Tsyvinski, A., 2014. Optimal taxes on fossil fuel in general equilibrium. *Econometrica* 82 (1), 41–88.
- Head, K., Mayer, T., 2014. Gravity Equations: Workhorse, Toolkit, and Cookbook, vol. 4, Elsevier, pp. 131–195.
- Hémous, D., 2016. The dynamic impact of unilateral environmental policies. *J. Int. Econ.* 103, 80–95.
- Hoel, M., 1996. Should a carbon tax be differentiated across sectors? *J. Public Econ.* 59 (1), 17–32, URL <https://www.sciencedirect.com/science/article/pii/0047272794014906>.
- Kortum, S., Weisbach, D., 2021. Optimal unilateral carbon policy. *SSRN Electron. J.*
- Kotlikoff, L., Kubler, F., Polbin, A., Scheidegger, S., 2024. Can today's and tomorrow's world uniformly gain from carbon taxation?. *Eur. Econom. Rev.* 168, 104819.
- Krusell, P., Smith, J., 2022. Climate Change Around the World. National Bureau of Economic Research, Working Paper 30338. URL <http://www.nber.org/papers/w30338>.
- Larch, M., Wanner, J., 2017. Carbon tariffs: An analysis of the trade, welfare, and emission effects. *J. Int. Econ.* 109, 195–213.
- Larch, M., Wanner, J., 2024. The consequences of non-participation in the Paris agreement. *Eur. Econ. Rev.* 163, 104699, URL <https://www.sciencedirect.com/science/article/pii/S001429212400028X>.
- Le Moigne, M., Lepot, S., Ossa, R., Ritel, M., Simon, D., 2024. Greening ricardo: environmental comparative advantage and the environmental gains from trade. WTO Staff Working Paper, World Trade Organization (WTO), Geneva, ERSD-2024-07, <https://hdl.handle.net/10419/306822>.
- Markusen, J.R., 1975. International externalities and optimal tax structures. *J. Int. Econ.* 5 (1), 15–29.
- Nordhaus, W.D., 1993. Optimal greenhouse-gas reductions and tax policy in the "DICE" model. *Am. Econ. Rev.* 83 (2), 313–317.
- Nordhaus, W., 2018. Projections and uncertainties about climate change in an era of minimal climate policies. *Am. Econ. J.: Econ. Policy* 10 (3), 333–360.
- OECD, 2022. Aggregate Trends of Climate Finance Provided and Mobilised by Developed Countries in 2013–2020. p. 25, URL <https://www.oecd-ilibrary.org/content/publication/d28f963c-en>.
- OECD, 2023. OECD inter-country input-output database. URL <http://oe.cd/icio>. (Accessed 15 April 2024).
- Parry, I.W., Black, S., Roaf, J., 2021. Proposal for an international carbon price floor among large emitters. *IMF Staff Climate Note*. URL <https://www.imf.org/en/Publications/staff-climate-notes/Issues/2021/06/15/Proposal-for-an-International-Carbon-Price-Floor-Among-Large-Emitters-460468>.
- Ricke, K., Drouet, L., Caldeira, K., Tavoni, M., 2018. Country-level social cost of carbon. *Nat. Clim. Chang.* 8 (10), 895–900.
- Ritchie, H., 2019. Who has contributed most to global CO2 emissions? Our World Data. <https://ourworldindata.org/contributed-most-global-co2>.
- Sandmo, A., 2006. Global public economics: Public goods and externalities. *Économie Publique/Public Econ.* 18–19 (1–2), URL <http://journals.openedition.org/economiepublique/4282>. Online since 17 October 2007, (Accessed on 13 May 2024).
- Shapiro, J.S., 2021. The Environmental Bias of Trade Policy. *Q. J. Econ.* 136 (2), 831–886.
- Stadler, K., Wood, R., Bulavskaya, T., Södersten, C.-J., Simas, M., Schmidt, S., Usubiaga, A., Acosta-Fernández, J., Kuenen, J., Bruckner, M., Giljum, S., Lutter, S., Merciai, S., Schmidt, J.H., Theurl, M.C., Plutzar, C., Kastner, T., Eisenmenger, N., Erb, K.-H., Koning, A., Deck, C.E.C.A.E., Rasul, K., Hertwich, E., Tukker, A., 2025. EXIOBASE 3. <http://dx.doi.org/10.5281/zenodo.15689391>.
- Tirole, J., 2016. Carbon pricing of a climate coalition. Unpublished manuscript.
- Weisbach, D., Kortum, S., Wang, M., Yao, Y., 2023. Trade, leakage, and the design of a carbon tax. *Environ. Energy Policy* 4, 43–90.